

**DOOMSCROLLING DETECTION AND DIGITAL MINDFULNESS
MOBILE APPLICATION FOR SHORT-FORM VIDEO PLATFORMS
USING VADER AND FUZZY LOGIC**

by

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Chapter 1

INTRODUCTION

Doomscrolling refers to the compulsive consumption of distressing or negative online content despite its adverse emotional and behavioral effects (Sharma et al., 2022). Mainstream device- and platform-level digital well-being tools still emphasize screen-time summaries, app timers, or break reminders instead of session-level interpretation of fast-changing feed content (Apple Inc., 2025; Google, 2024; Mosseri, 2021; Rahmillah et al., 2023; TikTok, 2024). This thesis addresses that gap through the design, development, and evaluation of the **Heuristic Risk-State Estimation System** — a privacy-preserving Android app that performs on-device risk estimation using behavioral signals, a text-first sentiment path, and an on-device Vision-Language Model (VLM) fallback for no-text items. Here, “detection” refers to computational risk estimation from observable proxies, not diagnostic classification.

The system combines threshold-based heuristics, interpretable fuzzy rule-based inference (Pickering et al., 2025; Vashishtha et al., 2023), a VADER-compatible text-first sentiment pipeline for low-resource code-mixed text (Hutto & Gilbert, 2014; Mohammed & Prasad, 2023; Nazir et al., 2026), and an on-device no-text VLM fallback implemented with Moon-dream 0.5B (Das & Singh, 2023; Sharshar et al., 2025). All processing occurs locally on the user’s device to preserve privacy under real-world mobile constraints. It is evaluated as a bounded software-engineering study, not a clinical trial, and focuses on practical usefulness, software quality, user acceptance, expert review, baseline convergent association, and short-term behavioral differences during a two-week field deployment.

1.1. Background of the Study

In recent years, doomscrolling has emerged as a behavioral concern in algorithm-driven social media environments. Short-form platforms such as TikTok, Facebook Reels, and Instagram Reels are built around rapid, recommendation-driven streams of content, and prior studies associate heavier or more compulsive use with attention-related strain, procrastination, reduced mindfulness, anxiety-related symptoms, and related negative outcomes (Canila et al., 2023; Cardoso et al., 2024; Hawwa et al., 2025; Qin et al., 2022; Taskin et al., 2024; Zhang, 2023).

Existing digital well-being tools address this problem only partly. Prior work shows that the emotional effects of scrolling depend on what users are exposed to, not just how long they scroll (Buchanan et al., 2021). Mainstream responses still center on time limits, usage summaries, or fixed break reminders, while app-level interventions reviewed in the literature remain heterogeneous and do not foreground privacy-by-design as a core requirement (Apple Inc., 2025; Google, 2024; Mosseri, 2021; Rahmillah et al., 2023; Tewari, 2023; TikTok, 2024). This suggests a narrower engineering gap for the present study: whether a private mobile tool can move beyond time alone by combining observable interaction patterns with a text-first sentiment proxy while still handling no-text items through an on-device visual fallback.

Mindfulness-oriented digital interventions provide limited support for brief app-delivered self-regulation prompts (Mitsea et al., 2023). Just-in-time smartphone interventions have also shown that smartphone-based support can be delivered in the moment, with feasibility and acceptability depending on timing, receptivity, and burden management (Mair et al., 2022; Roffarello & De Russis, 2021; Teepe et al., 2021; Yang et al., 2023). These works support in-the-moment prompting, but they do not show whether one system can continuously monitor short-form video behavior, process extracted text locally, resolve no-text

items through the VLM fallback, and issue adaptive prompts without cloud transmission or remote dashboards.

In the Philippine context, local peer-reviewed studies indicate that doomscrolling, problematic social-media use, and digital well-being are relevant concerns, although the strongest available evidence remains student-focused (Canila et al., 2023; Cardoso et al., 2024; Mike Jeizy P et al., 2024). Nearby adult Philippine contexts are more indirect, including social-media use among older adults and anxiety-related social-media exposure findings in adult workers (Castillo et al., 2022; Zamora et al., 2021). The local literature therefore supports contextual relevance, but not adult prevalence, validated thresholds, or expected intervention effects. Against that backdrop, the study examines whether the system can combine behavioral monitoring, a text-first sentiment proxy, the no-text VLM fallback, and adaptive prompting within a privacy-by-design architecture, then be evaluated for usability, reliability, and practical usefulness among adult Filipino users.

1.2. Statement of the Problem

This study addresses the gap left by time-based or platform-controlled digital well-being tools by asking whether a privacy-preserving Android app can be designed, developed, and evaluated for doomscrolling-related risk estimation on short-form video platforms. Current responses remain fragmented: platform-native tools are usually time-based, while related research analogues rarely combine local monitoring, content-sensitive estimation, and adaptive prompting in one deployable adult short-form video system. Because the available local literature does not yet support calibrated estimator thresholds or clinical interpretation, the problem is framed as a software-engineering question. Specifically, it seeks to answer the following questions:

1. What privacy-preserving mobile architecture and estimation framework can support doomscrolling-related risk estimation on the target short-form video

platforms using behavioral indicators and sentiment-related indicators when reliably resolvable, with 2-input behavioral fallback for sentiment-unreliable sessions?

2. What short-term Week 1-to-Week 2 changes are observed in selected logged usage metrics and self-reported doomscrolling scores between the intervention group and the logging-only control group, and within the intervention group across the same period?
3. What baseline convergent association exists between the fixed-prior Week 1 Doomscroll Severity Index (DSI) and participants' self-reported doomscrolling scores among participants with at least three sentiment-reliable Week 1 sessions?
4. How do users evaluate the system using the ISO/IEC 25010 software quality model and the Technology Acceptance Model (TAM)?
5. How do subject matter experts evaluate the system's technical design, privacy safeguards, heuristic logic, and intervention structure?

1.3. Significance of the Study

Contribution to the Body of Knowledge: This study contributes a mobile-systems design and pilot-scale field evaluation of one app that integrates behavioral monitoring, a text-first sentiment proxy, a no-text VLM fallback, adaptive prompting, and on-device privacy. Its contribution is an engineering evaluation framework for feasibility, acceptability, and estimator plausibility. The **Minimum Viable Lexicon (MVL)** workflow supports low-resource, code-mixed text handling within the system rather than functioning as a separate lexicon-development study.

Beyond its academic contribution, the study may offer value to the following stakeholders:

Adult users of short-form video platforms may benefit from a consent-based self-monitoring application that supports reflective interruption of problematic scrolling behavior without sending raw content to external servers.

Software developers and mobile-computing researchers may reuse the app’s architecture, hybrid text/VLM proxy design, and fallback logic when building privacy-preserving digital well-being systems for code-mixed social media environments.

Digital well-being researchers may use the study as an example of how behavioral logging, lightweight sentiment analysis, and adaptive prompting can be combined in a mobile field evaluation.

Educational institutions and digital well-being advocates may use the findings to support responsible discussions about privacy-preserving, user-directed self-regulation tools rather than surveillance-oriented monitoring systems.

1.4. Objectives of the Study

General Objective:

To design, develop, and evaluate the proposed system as a privacy-preserving, on-device Android application for doomscrolling-related risk estimation on short-form video platforms.

Specific Objectives:

1. To design and develop the system to estimate doomscrolling-related risk using behavioral indicators and sentiment-related indicators when reliably resolvable, with 2-input behavioral fallback for sentiment-unreliable sessions.
2. To determine the short-term Week 1-to-Week 2 changes in selected logged usage metrics and self-reported doomscrolling scores between the intervention group and the logging-only control group, and within the intervention group across the same period.

3. To determine the baseline convergent association between the fixed-prior Week 1 Doomscroll Severity Index (DSI) and participants' self-reported doomscrolling scores among participants with at least three sentiment-reliable Week 1 sessions.
4. To evaluate the system from the users' perspective using the ISO/IEC 25010 software quality model and TAM.
5. To obtain subject matter expert evaluation of the system's technical design, privacy safeguards, heuristic logic, and intervention structure.

1.5. Scope and Delimitations

This study covers the design, development, and evaluation of the system. The scope centers on the software artifact, its risk-estimation variables, and the two-week field evaluation.

Scope:

- Development of a native Android application using modern software engineering practices.
- Implementation and evaluation cover one deployable system and a pilot-scale two-week field study.
- Use of the Android Accessibility Service API for text extraction, interaction-event monitoring, and screenshot-assisted no-text routing through `AccessibilityService.takeScreenshot`, with related interaction signals used to estimate session duration, compute video dwell time from content transitions, and process sentiment-related indicators from visible text or no-text visual items. Swipe counts are retained only as supporting logs.
- Design of a modular architecture with separation between data collection, processing, and user interface components.

- Implementation of the core estimation framework using threshold-based rules, text-first VADER sentiment analysis, an on-device no-text VLM fallback using Moondream 0.5B, and fuzzy-logic inference, with unresolved cases handled through sentiment-unreliable classification and 2-input behavioral fallback.
- The system is designed for TikTok, Facebook Reels, and Instagram Reels as target platforms, although actual platform exposure during field deployment may vary by participant and empirical claims are limited to platforms that yield stable extraction.
- On-device processing, where raw text and temporary no-text screen frames are processed locally in RAM and discarded after scoring, while only aggregate local metrics and configuration data are retained on-device.
- A user-directed design in which risk estimates and prompts are delivered only to the consenting user on their own device, with no remote administrator dashboard.
- Evaluation uses a pilot-scale two-week deployment: all participants complete a one-week baseline logging phase, after which the intervention group receives adaptive prompts during Week 2 while the control group continues logging-only.
- Evaluation uses the ISO/IEC 25010 Software Quality Model (Functional Suitability, Performance Efficiency, Usability, Reliability), the Technology Acceptance Model (Perceived Usefulness, Perceived Ease of Use), subject matter expert review, and short-term observed differences under the study conditions.

Delimitations:

- The study targets Filipino Android users aged 18 years and above who actively use at least one of the target short-form video platforms. Its external validity does not extend to minors, iOS users, or users who primarily consume long-form or non-short-form content.

- Field deployment is limited to Android devices that can install the study build, maintain the required permissions, and keep the monitoring service sufficiently stable during the evaluation window.
- The risk-estimation framework addresses only short-form video platforms and does not cover long-form video, news websites, general web browsing, or other social media formats outside TikTok, Facebook Reels, and Instagram Reels.
- The study is not designed as a clinical diagnostic system, medical intervention, psychotherapy tool, or long-term habit-formation trial.
- The study does not claim to prevent doomscrolling. It evaluates a heuristic risk-estimation system and observes short-term behavioral differences between intervention and control groups.
- The study does not attempt multi-site rollout, large-sample validation, or extensive model optimization.
- The study excludes covert monitoring, employer surveillance, parental-control workflows, remote dashboards, and third-party alerting mechanisms.

1.6. Limitations

This study evaluates software behavior and short-term usage differences — not clinical outcomes. Changes in logged metrics should not be interpreted as reductions in anxiety, distress, or mental health conditions.

The following limitations describe constraints beyond the researchers’ control that may affect the study’s findings and generalizability:

- The use of Android Accessibility Service to read on-screen content from third-party applications may create platform-policy concerns. Ethical risk is reduced through explicit user consent, local processing, and the absence of third-party

surveillance workflows, but policy changes remain outside the researchers' control.

- The Accessibility Service approach is fragile by nature. Changes in the user interface of TikTok, Facebook, or Instagram may disrupt extraction logic, and Android device manufacturers may aggressively terminate background services, affecting data continuity.
- Continuous monitoring may affect battery life, thermal behavior, and background stability on some devices, which can in turn influence both system reliability and user acceptance.
- The system does not perform continuous full-resolution video or audio analysis. It relies on behavioral logs, accessibility-extracted text, and sparse transient on-device screen-frame sampling for no-text items. The system therefore still estimates risk from observable proxies rather than from full multimodal understanding of the viewed media.
- Although the deployable app supports Android 8.0 (API 26) and higher, the screenshot-assisted no-text VLM path depends on `AccessibilityService.takeScreenshot`, which is available only on Android 11 (API 30) and higher. On older compatible devices, no-text visual resolution degrades to the sentiment-unreliable / 2-input fallback path when screenshot capture is unavailable.
- Video duration is not consistently exposed across all target applications through the Accessibility Service. For this reason, the system relies on video dwell time and related interaction patterns rather than exact platform-reported video length.
- Because Android Accessibility APIs do not expose a guaranteed content-level identifier for third-party short-form feed items, transition detection remains

heuristic and may still mis-segment some consecutive items, especially when they are text-barren or visually similar.

- The baseline sentiment engine is English-centric. Although the system extends it with a limited Filipino/Taglish lexicon, unsupported dialects, slang drift, and code-mixed text may still reduce sentiment reliability.
- Sessions with high out-of-vocabulary text, or no-text items that still cannot be resolved reliably after VLM routing, are treated as sentiment-unreliable and excluded from some sentiment-dependent analyses, which may reduce the effective analyzable sample for some outcomes.
- Self-reported profile data and Doomscrolling Scale responses remain subject to recall bias and social desirability bias.
- Because recruitment is limited to consenting participants willing to install a monitoring application, the sample may overrepresent users who are already receptive to self-regulation tools.
- Although a logging-only control group is included to isolate prompt effects from monitoring awareness and natural fluctuation, the study remains non-blinded: intervention-group participants know they are receiving prompts, which may introduce expectancy or novelty effects beyond the prompt mechanism itself.
- For the intervention group, prompt-excluded active-use metrics can be mechanically lower by design when prompts occur because prompt-display time is removed from session duration and dwell-time calculations. For this reason, Chapter 3 treats raw elapsed session duration and raw elapsed dwell time as the primary behavioral comparison, while prompt-excluded metrics are kept only as supplementary traces of prompt-interrupted use.

- The two-week deployment window captures only short-term behavior. It is insufficient for demonstrating long-term habit formation, retention, or sustained behavioral change.
- Participants know they are being monitored, which introduces Hawthorne-effect risk and may reduce the naturalism of observed behavior.

1.7. Conceptual Framework

The study uses the **Input-Process-Output (IPO) model** to illustrate the implemented runtime architecture and data flow for heuristic risk estimation (Laudon & Laudon, 2022).

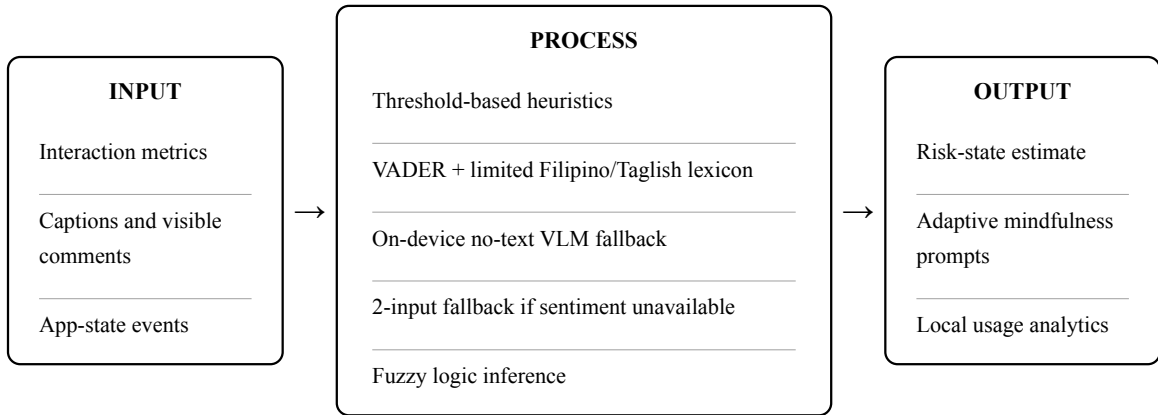


Figure 1: Conceptual Framework of the Study using IPO Model

The IPO model here refers to the runtime application pipeline rather than the full research workflow. During Week 2, a **feedback loop** operates once the live duration gate is met and the live RiskScore enters the Warning or Critical bands: the app selects an intervention level (L1 awareness, L2 pause, or L3 breathing exercise), closes the current interval, and begins a new one only if the user returns to the target platform. When sufficient reliable baseline sessions exist, the live prompt engine may personalize Session Duration and NSD memberships from Week 1 quantiles while Video Dwell Time remains fixed. Usable caption or visible-comment text stays on the text path; no-text items are routed

through `AccessibilityService.takeScreenshot` and the VLM fallback; unresolved cases degrade to 2-input behavioral inference with conservative L1/L2 prompts.

1.8. Theoretical Framework

The system is informed by the **Doomscrolling Feedback Loop Model** derived from Sharma et al. (2022). The model is used as a design guide for content exposure, prolonged engagement, and interruption points, but the study does not directly measure the internal emotional states described by the model.

The model conceptualizes doomscrolling as a three-phase cycle:

1. **Antecedents or triggers:** Users are drawn toward distressing or uncertainty-inducing content by factors such as anxiety, fear of missing out, or crisis-related information needs.
2. **Behavior:** Users engage in persistent, repetitive scrolling through negative or emotionally charged content.
3. **Outcome:** Exposure to such content contributes to distress or negative affect, which may in turn reinforce further scrolling.

This framework informs the system’s variable selection. The system estimates risk at the behavior and exposure level by observing video dwell time, session duration, and NSD as proxies for heightened maladaptive engagement. Swipe or scroll events are used only as transition signals to determine when one short-form item ends and another begins, allowing video dwell-time computation rather than serving as a separate estimator input. Its adaptive prompts are intended to interrupt potentially escalating patterns before exposure becomes more sustained.

The framework thus justifies using observable proxies and non-clinical prompts. Evaluation is guided by ISO/IEC 25010 and TAM, while the baseline-to-intervention design supports short-term behavioral comparison.

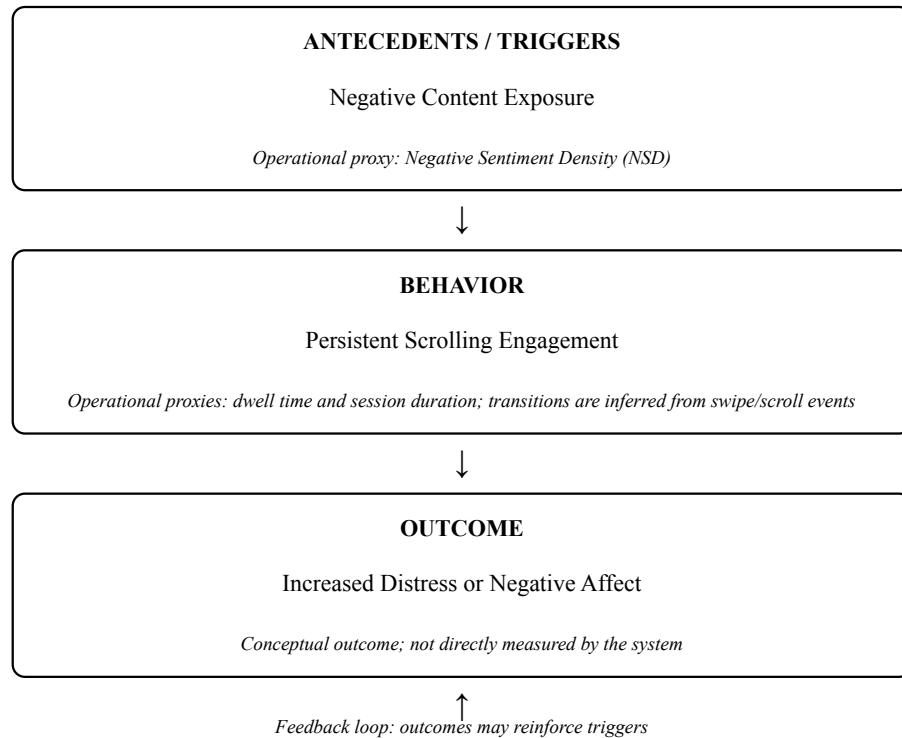


Figure 2: Doomscrolling Feedback Loop Diagrammatic Interception Model, adapted from Sharma et al. (2022)

The system is therefore a **heuristic risk-estimation tool** that uses behavioral and exposure proxies rather than direct measures of internal psychological states.

1.9. Definition of Terms

Adaptive Digital Mindfulness Prompts - Non-clinical system-generated interventions, such as awareness notifications, pause prompts, or short breathing exercises, that are triggered according to estimated risk severity to encourage reflective interruption of scrolling behavior.

Digital Mindfulness - A practice of intentional and self-aware technology use that promotes attention, presence, and self-regulation in digital environments (Aggarwal et al., 2024).

Doomscrolling - The compulsive and continuous consumption of negative or distressing content on digital platforms, particularly in ways that may reinforce anxiety, vigilance, or maladaptive engagement (Sharma et al., 2022).

Doomscrolling-Related Risk Estimation - The computational estimation of elevated scrolling risk using observable behavioral and sentiment-related proxies rather than direct measurement of internal psychological states.

Edge Computing - A computing approach in which data processing occurs on or near the data source - in this study, on the user's mobile device - rather than in remote cloud infrastructure, thereby reducing latency and privacy risks (Swathi et al., 2025).

Fuzzy Logic System - A computational approach that uses partial membership and rule-based reasoning to model ambiguous behavioral patterns, allowing graded risk estimation instead of binary classification through interpretable linguistic rules (Pickering et al., 2025; Vashishtha et al., 2023).

Heuristic Risk-State Estimation System - The Android application developed in this study for heuristic doomscrolling-related risk estimation and adaptive prompting.

ISO/IEC 25010 - An international software quality model used in this study to evaluate selected quality characteristics of the system, specifically Functional Suitability, Performance Efficiency, Usability, and Reliability (International Organization for Standardization, 2023).

Negative Sentiment Density (NSD) - The proportion of resolvable content units within a session that are classified as negative. In this study, it is computed from analyzable caption and visible-comment text units that pass the text-side reliability screen together with no-text items resolved through the VLM fallback. It is used as a limited proxy for negative exposure.

Technology Acceptance Model (TAM) - A framework used to assess user acceptance of digital systems, particularly through the constructs of Perceived Usefulness and Perceived

Ease of Use, and still commonly applied in mHealth acceptability evaluation (Adnan et al., 2025).

VADER Sentiment Analysis - Valence Aware Dictionary and sEntiment Reasoner, a lexicon- and rule-based sentiment analysis method for social media text that is adapted in this study for limited code-mixed use (Hutto & Gilbert, 2014).

Vision-Language Model (VLM) Fallback - A limited on-device visual-analysis path used in this study when a viewed short-form item has no usable caption or visible comments. It uses the Moondream 0.5B model together with `AccessibilityService.takeScreenshot` and transient RAM-only frame processing to estimate whether a no-text item contributes negative exposure.

Video Dwell Time - The time interval spent on a viewed short-form video before the user swipes away or otherwise transitions, computed from heuristically inferred content-transition signals derived from accessibility events and used in this study as one behavioral proxy for scrolling engagement.

Very Large Online Platforms (VLOPs) - Large digital platforms, such as Facebook, Instagram, and TikTok, that operate at massive user scale and strongly shape online content exposure and engagement patterns (Chen et al., 2024).

Session Duration - The continuous period of active engagement with a target short-form video platform, as determined by interaction signals and the session-management rules described in Chapter 3.

Sentiment-Unreliable Session - A session in which item-level negativity cannot be derived reliably for the main analysis after the available text path or no-text VLM path has been attempted, such as sessions dominated by high-OOV text or by unresolved no-text items. Such sessions are excluded from the primary NSD- and DSI-based analyses.

Swipe/Scroll Transition Signals - Accessibility-detected navigation events used to heuristically infer when one short-form video ends and another begins, with text verification

applied when usable caption text exists, enabling the computation of video dwell time and the logging of supporting interaction counts.

Chapter 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter reviews peer-reviewed literature and empirical studies relevant to doomscrolling, short-form platform engagement, digital mindfulness, privacy-preserving mobile monitoring, sentiment analysis, and software evaluation. Journalistic, policy, platform-issued, and product-documentation materials are used only for contextual background or feature verification. The review is organized to clarify the study’s evidence base, technical grounding, and remaining gaps.

2.1. Foreign Literatures

This section presents selected foreign literature relevant to doomscrolling, short-form platform design, digital mindfulness, and privacy-preserving mobile systems.

2.1.1. Doomscrolling, Platform Design, and Compulsive Engagement

Recent literature treats doomscrolling in short-form feeds as a joint product of user vulnerability and platform design rather than as a simple matter of screen-time volume. Rodrigues (2022) frames doomscrolling as compulsive information seeking that can intensify anxiety and helplessness, while Zenone et al. (2021) show that recommendation-driven environments can amplify harmful exposure and addictive-use patterns. Zhang (2023) complements that view by showing that TikTok is woven into users’ routines as entertainment, information, and social reference, which helps explain why these feeds can sustain engagement so effectively.

These works suggest that time-only intervention logic is too coarse for the present problem. A defensible intervention system should monitor sustained interaction patterns within algorithmic short-form feeds rather than rely only on coarse daily time limits.

2.1.2. Digital Mindfulness and Technology-Assisted Self-Regulation

Recent digital well-being reviews converge on a narrow design implication: technology-assisted self-regulation support is more acceptable when it is brief, timely, and low-burden. Mitsea et al. (2023) provide limited support for digitally assisted mindfulness as one self-regulation strategy, while Antezana et al. (2022) show that engagement improves when interventions remain lightweight and relevant to the user’s immediate context. Together, they support short adaptive prompts as low-burden, non-clinical interruption mechanisms.

2.1.3. Privacy, Persuasive Design, and Ethical Considerations in Digital Well-Being

Privacy-oriented digital well-being literature also constrains what a defensible system can do. Tewari (2023) argues that privacy protections should be designed into mobile health architectures from the start rather than added after collection logic is already fixed. That point is central to the present design: behavioral monitoring, sentiment processing, and prompting should occur without exporting raw user content to external services. This literature therefore supports local processing and aggregate-only retention.

2.1.4. Edge Computing and On-Device Vision-Language Fallbacks

Recent literature on privacy-preserving mobile analysis converges on a clear constraint: if visual inference is used at all, it should remain inside the device’s privacy boundary. Sending screenshots to cloud-based multimodal APIs would expose sensitive user data to external servers. Sharshar et al. (2025) support the feasibility of edge-deployed Vision-Language Models under network and latency constraints, while Lee et al. (2024)

show why compact transformer-based vision models remain a practical design priority on mobile and edge devices. Broader multimodal sentiment and affective-computing reviews likewise show that visual channels can add information when text is absent, although interpretation remains limited (Cortiñas-Lorenzo & Lacey, 2024; Das & Singh, 2023; Johnson et al., 2025). This literature supports the app’s on-device VLM fallback for no-text items, with Moondream 0.5B used as a small deployment-fit model.

2.2. Local Literatures

This section presents selected local literature and contextual materials relevant to doomscrolling, excessive social media use, and digital well-being in the Philippine setting. Because adult Philippine doomscrolling evidence remains limited, peer-reviewed local studies are used as nearby evidence of attention, scrolling, and self-regulation concerns, while journalistic or policy-oriented sources serve only as contextual background.

2.2.1. Local Discourse and Platform Concern

Philippine background materials show that doomscrolling and digital well-being had already entered local public discourse by 2021 Lanuza et al. (2021); Panaligan (2021). They establish contextual relevance, not adult behavioral prevalence, intervention effectiveness, or estimator validity.

2.2.2. Limits of the Local Evidence Base

The local empirical base remains uneven and still depends mainly on student-centered studies discussed later in this chapter. Recent Philippine work outside student-only samples is useful but indirect: Cleofas et al. (2022), Zamora et al. (2021), and Castillo et al. (2022) broaden the adult context by showing that social-media use intersects with mental well-being, anxiety-related susceptibility, or social connectedness across different Filipino

populations. Even so, the local literature does not yet provide adult doomscrolling field evidence, validated cutoffs, or intervention-effect benchmarks for a privacy-preserving mobile system.

2.3. Interim Synthesis of Literature

The reviewed foreign and local literature establishes three points. First, doomscrolling is a recognizable behavioral construct linked to distress-related and attention-related concerns, and platform design appears to help sustain engagement (Hawwa et al., 2025; Qin et al., 2022; Taskin et al., 2024). Second, existing digital well-being responses remain fragmented: mainstream device- and platform-level tools still emphasize summaries, timers, or break reminders rather than user-controlled, locally processed, content-sensitive estimation (Apple Inc., 2025; Google, 2024; Mosseri, 2021; Rahmillah et al., 2023; Tewari, 2023; TikTok, 2024). Third, Philippine literature supports local relevance and provides some adult context, but not enough adult field evidence to justify prevalence claims, threshold values, or expected intervention effects.

Together, these sources justify an engineering problem: combining behavioral signals, visible text, and a no-text VLM fallback in one privacy-preserving mobile system. They do not support text sentiment as a complete proxy for multimodal short-form feeds or provide strong local adult evidence for universal estimator parameters. Accordingly, the study uses a hybrid exposure signal in which analyzable text is scored through the text path and no-text items are routed through the VLM fallback.

2.4. Foreign Studies

This section presents a review of related studies that provide empirical evidence on doomscrolling, compulsive social media use, and their psychological and behavioral impacts.

2.4.1. Psychometric Validation of Doomscrolling as a Measurable Construct

Sharma et al. (2022) establish doomscrolling as a measurable self-report construct, while Satıcı et al. (2023) extend that psychometric work and support a shorter 4-item form. Together, these studies justify using self-reported doomscrolling as a comparison anchor, not as session-level ground truth. Chapter 3 therefore treats the scale as a baseline convergent-association reference rather than as a direct label for individual scrolling sessions.

2.4.2. Psychological Mechanisms and Consequences of Doomscrolling

Taskin et al. (2024) and Hawwa et al. (2025) both position doomscrolling as part of a wider pathway linking platform engagement to poorer well-being outcomes. Taskin et al. (2024) highlight reduced mindfulness and secondary traumatic stress as mediating mechanisms, while Hawwa et al. (2025) connect doomscrolling to the relationship between social media addiction and anxiety. Their shared implication is that a pure time-only measure is insufficient, yet neither study validates a single behavioral proxy or mobile risk-estimation rule by itself.

2.4.3. Platform Addiction and Algorithmic Engagement

Qin et al. (2022) show that TikTok’s platform characteristics can intensify flow experience and addictive use. This supports the thesis’s focus on short-form mechanics and sustained engagement rather than on user weakness alone.

2.4.4. Computational Approaches to Detecting Harmful Digital Behavior

Zannettou et al. (2024) show that engagement with TikTok recommendation streams can be studied from donated behavioral traces, while Lokeshkumar et al. (2021) show that text and activity patterns can be processed computationally as risk-relevant indicators. These studies serve as methodological analogies: they support technical plausibility for

trace-based analysis, but not direct precedent for doomscrolling estimation on short-form platforms or validation of the specific proxies and privacy constraints used here.

2.4.5. Fuzzy Logic as an Inference Engine for Mobile Behavioral Classification

Recent reviews show that fuzzy rule-based systems are useful when the target construct is gradual rather than binary and when the inference process must remain interpretable to human reviewers (Pickering et al., 2025; Vashishtha et al., 2023). Recent work on interpretable fuzzy-model design also continues to treat simple triangular or shoulder-style membership sets as practical, auditable choices when labeled session-level training data are not available (Khairuddin et al., 2021; Porebski, 2022). Recent fuzzy-design work also describes symmetric grid partitions and strong fuzzy partitions as interpretable starting structures when domain-specific numeric cutoffs are unavailable (Azam et al., 2021; Casalino et al., 2022; Khairuddin et al., 2021). Together, this literature supports using fuzzy logic and an interpretable starting structure for Negative Sentiment Density, where no universal percentage threshold set is established.

2.4.6. Data Extraction and Privacy-Preserving Edge Computing

Lee et al. (2022) show that Android Accessibility Service can expose fine-grained interaction data unavailable through standard APIs, but they also make clear that such access is privacy-sensitive. Swathi et al. (2025) complement that point by supporting edge-centric monitoring as a privacy-preserving and responsive architecture. Together, these studies justify the architectural stance that rich interaction data, if collected at all, should be processed locally and reduced to aggregate outputs rather than exported as raw content.

2.4.7. Social Media Use, Sustained Attention, and Digital Well-Being Interventions

Rajeswari & Vakkil (2024) and Roffarello & De Russis (2021) support different parts of the present design. Rajeswari & Vakkil (2024) link heavier social media engagement with poorer sustained attention, which supports the general relevance of prolonged use and viewing cadence as behavioral signals. Roffarello & De Russis (2021) show that smartphone habits can be monitored in real time and mitigated through proactive just-in-time reminders, which supports the feasibility of lightweight adaptive intervention on the device. Together, they justify attention to both continuous engagement patterns and in-the-moment interruption, though not direct validation of the present doomscrolling estimator. Recent measurement studies also help bound the fuzzy time-based inputs, even though they do not validate doomscrolling-specific cutoffs. Muise et al. (2024) show that smartphone content exposure often occurs in bursts of only a few seconds, Cho et al. (2021) treat sessions shorter than 5 seconds as likely mistaken openings and use 30 seconds as a short-duration divider in feature-level social-media logging, and Tian et al. (2021) report that 93.7% of mobile app usage in their large-scale logs lasts less than 10 minutes. Recent smartphone-logging work that reconstructs sessions from app-event streams also continues to merge adjacent app uses into one session when inter-app gaps remain within about 45 seconds (Ahmed & Ahmed, 2023; Chen et al., 2023). These studies justify distinguishing immediate skips, brief but nontrivial viewing, clearly sustained use, and very short app-switch interruptions instead of treating all exposure as one undifferentiated time scale. Intervention-timing studies point in the same direction. Terzimehić & Aragon-Hahner (2022) report that sessions shorter than 10 minutes are less prone to regret, with alternative offline activities becoming more salient in the 10-20-minute and >20-minute ranges. Rixen et al. (2023) treat 10+ minute infinite-scrolling episodes as long sessions, Meinhardt et al. (2025) deploy an infinite-scroll intervention after 15 minutes of continuous scrolling, Ismail & Al Thani (2022) use personalized prompts after 40 minutes of continuous sitting, and

Ikegaya et al. (2025) show that week-1-derived personalized intervention criteria outperform uniform criteria in the first hour after prompting. Recent JITAI reviews add the same design implication at the framework level: decision points and rules should be empirically grounded, burden-aware, and personalized when possible rather than fixed by one universal cutoff (Fiedler et al., 2024; Genugten et al., 2025; Hsu et al., 2025). Together, these studies support treating sub-10-minute use as a lower-risk baseline region, recognizing 15 minutes as a defensible live-prompt entry point for continuous infinite scrolling, and deriving Week 2 prompt criteria from each participant’s Week 1 baseline.

2.4.8. Technology Acceptance Frameworks

Recent mHealth acceptability reviews continue to treat TAM as a practical evaluation framework, with Perceived Usefulness and Perceived Ease of Use retained as core acceptance constructs (Adnan et al., 2025). That supports the inclusion of TAM alongside ISO/IEC 25010 in the post-usage evaluation.

2.4.9. Computational Constraints in Code-Mixed Sentiment Analysis

Analyzing Taglish on mobile devices presents two challenges: computational load and high Out-of-Vocabulary (OOV) rates. Recent low-resource and code-mixed sentiment work shows that transformer approaches can be data-hungry or resource-intensive, while lexicon-based and hybrid methods remain relevant when reproducibility, low overhead, and resource scarcity matter (Hussain et al., 2025; Mohammed & Prasad, 2023; Nazir et al., 2026; Tho et al., 2021). Work on lexicon curation for low-resource social-media text also shows that slang, emoticons, and local usage patterns often require targeted valence tuning rather than direct reuse of generic sentiment resources (Hashmi et al., 2024; Khan et al., 2025; Perera & Caldera, 2024; Wijayanti & Arisal, 2021). For that reason, the study adopts

a limited VADER + **Minimum Viable Lexicon** (MVL) strategy for code-mixed sentiment analysis on mobile devices.

2.4.10. Addressing the Affective Gap Through an On-Device VLM Fallback

When a short-form item contains no usable caption or visible comments, text-only sentiment cannot characterize that item at all. Reviews of multimodal sentiment analysis show why visual channels become relevant once text is absent (Das & Singh, 2023), while user-generated-video emotion studies show that keyframe-based reduction can support practical visual inference from video without requiring dense frame-by-frame processing (Wei et al., 2021; Zhang & Xu, 2023). Additional recent work shows that sparse visual sampling can remain useful under resource or privacy constraints, including sparse sampling with majority voting on short utterance clips (Sharma et al., 2023), privacy-compliant group emotion recognition using only 5 uniformly distributed frames on 5-second videos (Augusma et al., 2023), and sparse sampling regimes in recent VLM long-video evaluation under context-length constraints (Qu et al., 2025). These studies justify routing no-text items through the VLM fallback, using Moondream 0.5B as a small deployment-fit model with an on-demand screenshot capture rule rather than continuous video analysis. They do not validate the exact prompt design or label set, and they do not support full multimodal affective understanding.

2.5. Local Studies

2.5.1. Doomscrolling and Student Outcomes in the Philippines

Mike Jeizy P et al. (2024) provide the closest local study to the present topic by describing doomscrolling-related experiences among Filipino students. Bautista et al. (2024), Canila et al. (2023), Ababat et al. (2024), and Cleofas et al. (2022) reinforce the

same local signal: heavier or more problematic social media use is linked to boredom proneness, distraction, procrastination, attention-related strain, or poorer mental well-being among Filipino students and young adults. These studies establish Philippine relevance but remain student-centered, self-report-heavy, or qualitative, so they support cautious adult field evaluation rather than adult norms or threshold values.

2.5.2. Social Media, Reading Competence, and Attention Span

Gagalang (2021) and Cardoso et al. (2024) together support the local relevance of attention-related harms associated with heavy digital-platform use, especially among students. They help explain why viewing cadence and sustained use are reasonable behavioral dimensions to observe, even though they do not validate the estimator thresholds used for an adult sample.

2.5.3. Cognitive Benefits of Digital Detox Interventions

Lim et al. (2025) show that a two-week digital detox can improve attention and memory among Filipino Grade 12 students. The study is relevant mainly because it makes a two-week observation window plausible, not because it is a direct precedent for the present prompt-based intervention.

2.5.4. Local Computational Work on Filipino Sentiment Analysis

Co et al. (2022) and Cruz & Balahadia (2022) show that Filipino or English-Filipino public text can be analyzed with lexicon-based or lightweight sentiment pipelines, but they also make clear that domain adaptation remains necessary when language use shifts across settings. These studies support the narrower claim that Filipino-facing sentiment processing is technically workable in the local setting, provided the lexicon is treated as task-specific and incomplete rather than as a universal sentiment resource.

2.5.5. Software Quality Evaluation in Philippine Educational Technology

For software evaluation, the stronger methodological anchor is the official ISO/IEC 25010 product quality model itself rather than a single local implementation paper (International Organization for Standardization, 2023). The standard supports the choice of selected quality characteristics, while the survey wording used in Chapter 3 remains researcher-developed and subject to expert review.

2.6. Integrated Synthesis and Research Gap

2.6.1. Comparative Analysis of Existing Digital Well-Being Tools

To contextualize the research gap identified in this review, the following table summarizes the capabilities of existing digital well-being tools and platform-level features. The comparison is based on five key requirements derived from the preceding literature. Feature characterizations for Apple Screen Time, Google Digital Wellbeing, TikTok, and Instagram are drawn from their official product or support pages (Apple Inc., 2025; Google, 2024; Mosseri, 2021; TikTok, 2024). The broader third-party app row is grounded in the multimethod review by Rahmillah et al. (2023).

Operational Definitions for Comparison:

- **Behavioral Monitoring:** [Real-time] = continuous background monitoring of specific actions; [Aggregate] = summary views or timer-based monitoring without item-level sensing; [Platform-only] = monitoring limited to one platform's own activity tools.
- **Text-Visible Sentiment Proxy:** [✓] = analyzes the emotional tone of extracted text content; [×] = treats all viewed content equally regardless of text tone.
- **Adaptive Intervention:** [✓] = intervention type changes based on estimated severity; [Partial] = fixed caps, timers, or break nudges that do not estimate

content or risk; [×] = monitoring only or fixed reminders without severity adaptation.

- **On-Device Processing:** [✓] = core monitoring or intervention logic is user-controlled and executed locally on the device; [N/A] = feature is platform-managed and not exposed as a user-controlled local processing pipeline.
- **Platform Scope:** [System-wide] = works across many apps on the device; [Single-platform] = limited to one service; [Varies] = depends on the specific third-party app; [Specific targets] = limited to named study platforms.

Table 1. Comparative Analysis of Digital Well-Being Tools and the Proposed System

Tool / System	Behavioral Monitoring	Text-Visible Sentiment Proxy	Adaptive Intervention	On-Device Processing	Platform Scope
Apple Screen Time	Aggregate	×	Partial	✓	System-wide
Google Digital Wellbeing	Aggregate	×	Partial	✓	System-wide
TikTok screen-time tools	Platform-only	×	Partial	N/A	Single-platform: TikTok
Instagram daily limit / break reminders	Platform-only	×	×	N/A	Single-platform: Instagram
Rahmillah et al. (2023) apps	Varies	×	Partial	Varies	Varies
Proposed System	Real-time	✓	✓	✓	Specific targets: TikTok, Facebook Reels, Instagram Reels

As shown in the table, the reviewed literature and official product pages identify useful summaries, timers, and break mechanisms, but not a single adult short-form video system that combines real-time behavioral monitoring, a text-first sentiment proxy, an implemented no-text VLM fallback, adaptive prompting, and strict local processing. The gap therefore lies in integration and deployment scope: the ingredients appear separately in prior work, but not as one privacy-preserving field-tested artifact under the conditions targeted here. Buchanan et al. (2021) help motivate this search by showing that scrolling outcomes depend on exposure quality rather than time alone, while Roffarello & De Russis (2021) show

that just-in-time mobile interventions are feasible even without the specific doomscrolling construct or item-level negativity proxy used here.

The reviewed studies converge on three themes. First, doomscrolling can be measured through self-report and discussed as a distinct construct, which supports a baseline convergent association check (Satici et al., 2023; Sharma et al., 2022). Second, technical and intervention precedents exist for mixed mobile interventions, interpretable fuzzy inference, privacy-preserving on-device processing, Filipino-facing sentiment analysis, and narrow no-text visual inference on edge devices, although the closest studies remain analogies rather than direct implementations of the app evaluated here (Co et al., 2022; Cruz & Balahadia, 2022; Das & Singh, 2023; Lee et al., 2022; Pickering et al., 2025; Roffarello & De Russis, 2021; Sharshar et al., 2025; Vashishtha et al., 2023; Wei et al., 2021). Third, the local Philippine literature indicates that attention, self-regulation, and problematic-use concerns are locally relevant, that some adult social-media contexts have been studied, that short-term digital well-being observation windows are workable, and that Filipino-facing sentiment processing is technically feasible in local text settings (Castillo et al., 2022; Cleofas et al., 2022; Co et al., 2022; Cruz & Balahadia, 2022; Lim et al., 2025; Mike Jeizy P et al., 2024; Zamora et al., 2021). It does not yet establish adult prevalence, intervention effects, or calibrated threshold values for the present estimator.

A clear **research gap** therefore remains. No reviewed study integrates behavioral logging, a text-first sentiment proxy, an implemented no-text VLM fallback, adaptive prompts, and strict local processing in one privacy-preserving adult short-form video system while also reporting a field evaluation aligned with this thesis. The local adult evidence base is also too limited to justify calibrated estimator thresholds, strong expected-effect claims, or clinical interpretation. Hybrid text-plus-visual sentiment routing is therefore treated as partial coverage of negative exposure, not as a full representation of multimodal short-form video

content. In this study, NSD is defined as a session-level negative-exposure metric computed from resolvable text units and VLM-resolved no-text items. The study addresses these gaps through a two-week pilot field evaluation focused on software behavior, short-term behavioral comparison, user acceptance, expert review, baseline convergent association, and sentiment-reliable coverage reporting.

Chapter 3

METHODOLOGY

This chapter presents the methodology used to design, develop, and evaluate the system. It covers the research design, architecture, development procedure, evaluation method, data collection, respondents, and statistical treatment. Here, the term “detection” refers to heuristic risk-state estimation from observable proxies rather than diagnostic classification.

3.1. Type of Research

This study adopts a **design-and-development research** approach with a **two-group baseline-intervention pilot field evaluation** (Creswell & Creswell, 2022). The design-and-development component focuses on building the software artifact and its heuristic risk-estimation logic, while the field-evaluation component observes the system under real-world conditions.

The study follows a primarily quantitative orientation, with open-ended feedback used as supplementary qualitative context. Automated system logs and structured survey responses are used to assess software quality, user acceptance, expert review, baseline convergent association, and short-term behavioral differences. The study is a software evaluation rather than a clinical trial, and findings are interpreted within the study conditions.

3.2. Research Design

The study uses a **two-group baseline-intervention field design** treated as a **pilot-scale randomized field evaluation**. After consent and study-compatibility screening,

participants are assigned 1:1 to either an **intervention group** or a **logging-only control group** using a concealed permuted-block allocation list with randomly ordered block sizes of 2 and 4. The allocation list is generated before enrollment by a researcher who is not responsible for participant screening and is released only after eligibility is confirmed, which helps preserve balance without making the next assignment easily predictable. All participants then complete an identical one-week baseline logging phase with prompts disabled. During Week 2, the intervention group receives adaptive prompts while the control group continues logging-only.

The study questions are answered through related but distinct methodological parts. Research Question 1 is addressed by the architecture, routing, variable-justification, and algorithm sections. Research Question 2 is addressed by Week 1-to-Week 2 change analyses of selected logged metrics and self-reported doomscrolling scores, comparing the intervention and control groups and using a paired within-intervention comparison as secondary evidence. As a pilot-scale study, these analyses estimate short-term observed differences under the study conditions rather than definitive intervention efficacy. Research Question 3 is addressed by the baseline convergent association between Week 1 DSI and the Doomscrolling Scale. Research Questions 4 and 5 are addressed by the ISO/IEC 25010, TAM, and SME evaluations.

3.3. System Architecture

The system uses a **modular three-layer architecture** to support maintainability, interpretability, and separation of concerns. Its design follows **Privacy by Design**, keeping processing local to the device through an edge-computing approach. The application is organized into the **Data Access Layer**, **Business Logic Layer**, and **Presentation Layer**, allowing data collection, estimation logic, and user-facing components to be developed

with minimal coupling. The architecture also avoids device rooting by relying on Android Accessibility Service for text extraction, interaction-event monitoring, and screenshot-assisted VLM fallback when no usable caption or visible comments exist. The figure below shows this hybrid text/VLM pipeline.

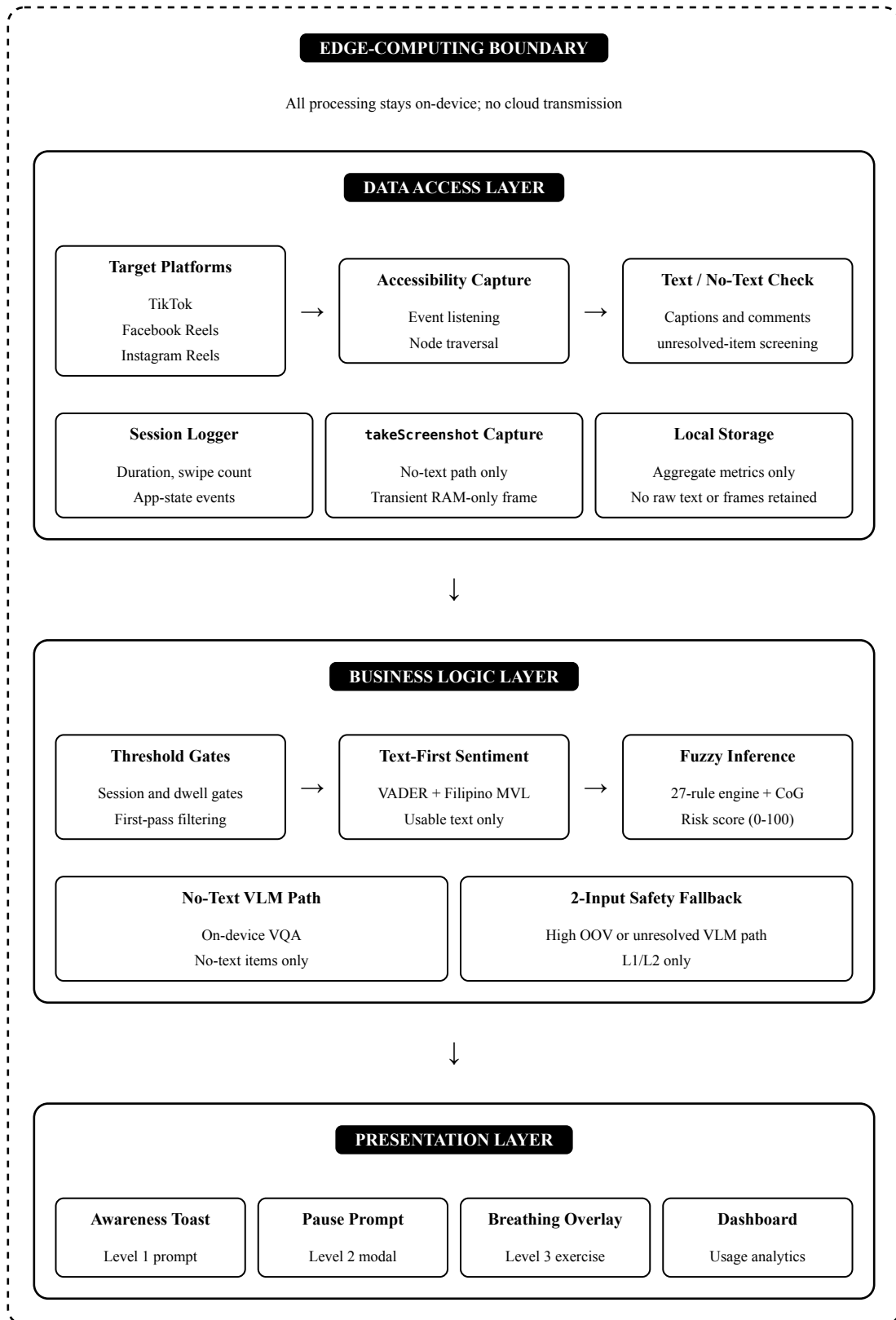


Figure 3: System Architecture Diagram: Three-Layer Edge Computing Design

3.3.1. Data Extraction Strategy

The primary mechanism for data acquisition is the Android **Accessibility Service API**. It lets the system read captions, comments, and interaction events from short-form video apps without requiring device rooting. The same layer also captures the interaction signals needed for session duration, content-transition detection, video dwell-time computation, and supporting swipe logs. When a viewed item has no usable caption or visible comments, the enabled accessibility service uses `AccessibilityService.takeScreenshot` to capture an on-demand transient screen frame for the no-text VLM fallback.

3.3.2. Accessibility Service Implementation

The implementation entails creating a background service that extends the `AccessibilityService` class. Functionally, this allows the system to observe the UI hierarchy of other apps in real-time, effectively perceiving the screen content as a screen reader would for accessibility purposes.

The technical implementation involves the following core components:

- **Service Extension:** Development of a custom service inheriting from `AccessibilityService`.
- **Event Listening:** The service subscribes to specific system events, primarily `AccessibilityEvent.TYPE_WINDOW_CONTENT_CHANGED` and `TYPE_VIEW_SCROLLED`, to detect dynamic content updates.
- **Node Traversal:** Upon capturing an event, the system traverses the hierarchy of `AccessibilityNodeInfo` objects, filtering specifically for `TextView` elements to extract creator captions and any visible user comments already present on-screen as affective reaction signals. The system does not actively open hidden comment panels or trigger additional UI states during extraction.

- **Scroll-Event Disambiguation:** Because `TYPE_VIEW_SCROLLED` events are generated by both feed-level video transitions and within-video comment-panel scrolling, the system does not treat every scroll event as a new item by default. When usable caption text is present, the system applies a content-change verification step before registering a content transition: after each feed-level scroll event, the caption text is re-extracted and compared to the previously recorded caption using exact string equality after whitespace normalization, and a transition is registered only if the caption has changed. When a viewed item has no usable caption or visible comments, however, the system cannot rely on caption comparison; in that no-text case, a feed-level scroll event is treated as a heuristic new-item boundary unless suppressed by comment-panel state or similar non-feed interaction context. As a secondary check, the system tracks comment-panel open/close state via `TYPE_WINDOW_STATE_CHANGED` to suppress transition processing during comment browsing. Comment-panel scrolling, caption expansion, and UI-only animations therefore do not reset the dwell-time interval or increment the swipe count. Comments visible during panel scrolling are still extracted for sentiment analysis and associated with the current video item.

3.3.3. Constraints and Challenges

A significant challenge in this architecture is the requirement to reverse-engineer the UI structure of targeted applications. The extraction logic must identify specific resource IDs (e.g., `com.zhiliaoapp.musically:id/caption_text`) to distinguish meaningful content from UI clutter such as buttons or timestamps. This approach has inherent fragility: changes to the target application’s UI structure or IDs by their respective developers may disrupt the scraper’s functionality, representing a known limitation of this methodology.

3.3.4. Edge Computing and Privacy Architecture

The system follows edge computing principles to mitigate privacy risks:

1. **No Persistent Storage of Raw Content:** In the study version, scraped text content and temporary no-text screen frames are processed entirely in RAM and immediately discarded after scoring. Only derived numeric scores, aggregate usage metrics, and local configuration data are retained on-device.
2. **On-Device Processing:** All Fuzzy Logic inference, VADER analysis, and no-text VLM inference occur locally on the user’s device. No content is transmitted to external servers.
3. **User-Initiated Consent:** The Accessibility Service requires explicit user activation through Android system settings. The same enabled service provides the no-text screenshot capability through `android:canTakeScreenshot="true"` and `AccessibilityService.takeScreenshot`, so the study’s operational data-access consent is centered on the Accessibility Service setup.
4. **Aggregate-Only History:** The usage dashboard displays only aggregate statistics (e.g., “negative content exposure: 45%”) rather than storing individual text samples or captured frames.

3.3.5. Extraction Failure Handling

This architecture relies on hardcoded UI resource IDs (e.g., `com.zhiliaapp.musically:id/caption_text`) within the Accessibility Service. Mid-experiment UI updates can restructure the target application’s view hierarchy and disrupt extraction. Such breakage is logged as a **reliability event**, and sessions or participant-periods that cannot yield valid extraction output are excluded from the analyses that require those measures.

3.3.6. Mindfulness Intervention System

Based on digital well-being and smartphone-habit intervention literature, including recent just-in-time adaptive intervention work on timing, feasibility, and burden management (Ismail & Al Thani, 2022; Mair et al., 2022; Roffarello & De Russis, 2021; Teepe et al., 2021; Wang et al., 2023; Yang et al., 2023), the system uses a multi-level adaptive intervention approach:

Level 1 - Awareness Notification: A non-intrusive toast message appears when the RiskScore enters the lower Warning band, informing the user of their session duration.

Level 2 - Pause Prompt: A modal dialog offers the user options to continue, take a break, or view their usage statistics when the RiskScore enters the upper Warning band.

Level 3 - Breathing Exercise: A full-screen overlay presents a brief guided breathing animation (study default: 60 seconds) when the RiskScore enters the Critical band. The breathing overlay is treated as a study-defined pause-and-reset prompt rather than as a therapeutic procedure (Mitsea et al., 2023; Yang et al., 2023). It is intended as behavioral interruption only.

For analytic transparency, time spent inside system-generated modals or overlays is excluded from prompt-excluded active-use metrics. When an intervention appears, the ongoing content-view interval is closed before the prompt begins, and a new interval starts only if the user returns to the target platform afterward. Raw elapsed session time and raw elapsed dwell time are retained separately for the primary behavioral comparison in the evaluation phase.

Usage Dashboard: Shows aggregated usage and sentiment trends over time to support self-reflection and behavioral awareness. It is treated as a secondary reflective feature rather than as the primary intervention mechanism.

3.3.7. Design Justification for Risk-Estimation Variables

This section addresses Research Question 1 by mapping each risk-estimation variable to the Doomscrolling Feedback Loop Model (Sharma et al., 2022) and to the reviewed literature. Variables were retained only if they were conceptually plausible, literature-supported, and observable through Android without rooting.

Table 2. Design Justification Matrix: Theoretical Constructs to System Variables

Theoretical Construct	Observable Proxy	System Variable	Data Source	Unit
Behavior: Persistent, obsessive consumption	Prolonged or rapid navigation through content without meaningful pauses	Video Dwell Time	AccessibilityService (active time between TYPE_VIEW_SCROLLED events, gated by micro-interaction events)	seconds
Behavior: Repetitive rapid feed navigation	Frequency of moving through items during a session	Swipe Count (supporting interaction log)	AccessibilityService view-scroll event stream	count
Behavior: Prolonged maladaptive engagement	Extended time on target apps without voluntary breaks	Session Duration	Accessibility-derived session tracker and app-state events	minutes
Antecedents/ Triggers: Negative content exposure	Proportion of consumed content with negative emotional tone	Negative Sentiment Density (NSD)	Negative-exposure density from analyzable text units and VLM-resolved no-text items	% (0–100)

Video Dwell Time was selected because TikTok and Reels use a discrete, paginated swipe interface rather than continuous scrolling. Dwell time - measured as the duration in seconds between consecutive content-verified transition events — captures how quickly or slowly users move through content. Because the Accessibility Service cannot reliably read exact platform-reported video duration across all target applications, dwell time is treated as a behavioral proxy. **Swipe Count** is retained only as a supporting interaction metric that helps show whether a long session consists of rapid chaining or slower browsing.

Session Duration is operationalized as continuous active time on a target application, reflecting the prolonged-engagement dimension reported in doomscrolling-related literature. Together, dwell time and session duration describe how long and how continuously the user

remains engaged. They are retained as observable engagement proxies, not as validated biomarkers of doomscrolling.

Negative Sentiment Density (NSD) captures the negative-exposure dimension of doomscrolling through a hybrid operationalization. When usable text exists, analyzable caption and visible-comment units are scored through the text path. When no usable text exists, the item is routed through the VLM fallback and contributes one resolved no-text label to the same session-level exposure estimate. High-OOV text is not rerouted into the VLM; if negativity still cannot be resolved reliably, the session enters the 2-input behavioral fallback and is marked sentiment-unreliable for the main sentiment-based analyses.

NSD is therefore an exposure proxy, not a direct measure of distress. It cannot distinguish crisis-related negativity from ordinary negative commentary or represent the full multi-modal meaning of viewed media.

Session Boundary Definition: A session begins when a target app enters the foreground (detected via `TYPE_WINDOW_STATE_CHANGED`) and ends when: (a) the target app leaves the foreground for longer than a 30-second bridge window, (b) the screen turns off, or (c) the Accessibility Service is terminated. If the user briefly switches away and returns within 30 seconds, the gap is excluded from duration and dwell-time accumulation but the session continues. Recent smartphone-logging studies that reconstruct sessions from app-event streams continue to use 45-second inter-app gaps when direct lock/unlock boundaries are unavailable (Ahmed & Ahmed, 2023; Chen et al., 2023). The present system therefore treats 30 seconds as a stricter study-defined session-continuity heuristic within that recent measurement range so only very brief app switches are merged; longer away intervals are treated as new sessions. The post-deployment sensitivity analysis examines ± 15 -second perturbations alongside the other fixed study priors.

Idle Timeout Heuristic (Active-vs-Idle Disambiguation): To prevent false inflation of dwell time when the device is unattended (e.g., phone left on a looping video), the dwell

accumulator is treated as **active** only when periodic micro-interactions are observed (e.g., TYPE_VIEW_CLICKED, TYPE_TOUCH_INTERACTION_START/END, or equivalent interaction signals exposed by the Accessibility layer). If no micro-interaction occurs within a rolling 45-second window, the ongoing dwell interval is paused and marked as idle exposure until interaction resumes or the next swipe occurs. Recent smartphone-session reconstruction studies continue to treat gaps of about 45 seconds as a plausible continuity delimiter when direct boundary data are unavailable (Ahmed & Ahmed, 2023; Chen et al., 2023). The dwell tracker uses the same order-of-magnitude value as a study-defined inactivity cap to distinguish active viewing from unattended looping: if interaction signals disappear beyond that range, accumulation pauses until interaction resumes or the next swipe occurs.

$$\text{NSD} = \left(\frac{\text{Negative resolvable units}}{\text{Resolvable units in session}} \right) \times 100$$

To formally represent this calculation in mathematical notation:

$$\text{NSD} = \frac{\sum_{i=1}^{N_R} \mathbb{I}(y_i = 1)}{N_R} \times 100$$

Where N_R is the number of session units whose negativity can be resolved either from analyzable text or from the no-text VLM path, and y_i is the binary negative label assigned to resolved unit i . For text-resolved units, negativity follows the VADER criterion $C_i < -0.05$. For no-text VLM-resolved units, negativity is assigned when the returned visual label is SEVERE_NEG or MILD_NEG. The indicator function $\mathbb{I}$ assigns 1 only to resolved units classified as negative and 0 otherwise. This makes the reported NSD a conservative negative-item density rather than a continuous magnitude-weighted affect score.

The -0.05 threshold follows VADER’s standard classification convention, where compound scores below -0.05 are treated as negative sentiment (Hutto & Gilbert, 2014). Caption/comment text units excluded by the language-reliability screen are removed from the text-side evaluation. No-text items routed to the VLM contribute one resolved item-level label each: SEVERE_NEG and MILD_NEG count as negative exposure, while NEUTRAL,

MILD_POS, and SEVERE_POS count as resolved non-negative exposure. High-OOV text is not rerouted into the VLM path. Sessions with very few resolvable units are flagged descriptively during analysis.

Alternative candidates such as swipe-back frequency, time-of-day patterns, and biometric signals were excluded because they either lack enough empirical support in the doomscrolling literature or are technically infeasible within the intended Android monitoring environment. Likewise, although the **internal psychological Antecedents** of the Feedback Loop Model (e.g., anxiety, FoMO) remain conceptually important, they are not measured directly because real-time capture would require intrusive experience sampling or biometric sensors. The system therefore estimates doomscrolling-related risk from one exposure proxy, NSD, together with two behavioral proxies, Video Dwell Time and Session Duration, while swipe or scroll events remain transition signals and supporting logs rather than a separate estimator input.

3.3.8. Analytic Scoring Framework

Week-level DSI is computed on one fixed scoring framework across both study weeks so the analytic scale remains constant. Swipe events serve only as transition markers for dwell-time computation, and prompt-response logs are reported descriptively because they exist only in Week 2. Live prompting may personalize Session Duration and NSD memberships after Week 1, but analytic DSI scoring remains fixed across both weeks.

3.4. Hardware and Software Specifications

Development uses Android Studio (Ladybug 2024.2.1+) with Kotlin 1.9+ on a machine with at least 8 GB RAM. The deployable application targets Android 8.0 (API 26) or higher, while the screenshot-assisted no-text VLM path requires Android 11 (API 30) or higher because it uses `AccessibilityService.takeScreenshot`. Local storage uses

Room (SQLite), sentiment analysis uses a VADER-compatible lexicon-and-rule engine re-implemented in Kotlin for on-device Android deployment, dashboard visualization uses MPAndroidChart, screen-content extraction uses `androidx.accessibilityservice`, and no-text routing uses the Android Accessibility Service together with the on-device Moon-dream 0.5B VLM/VQA module. A physical Android device is used for field testing.

3.5. Method in Developing Software

3.5.1. Methodology Procedure

3.5.1.1. Agile Scrum

Development follows **Agile Scrum** in four iterative one-week sprints covering: (1) requirements finalization and planning, (2) accessibility extraction and local logging, (3) estimator and prompting integration, and (4) testing, refinement, and deployment preparation. Scrum is selected because the Accessibility Service extraction logic requires frequent adaptation as target-app UI structures change, and the fuzzy-logic rule base requires iterative tuning.

3.5.2. Algorithm

The core logic of the system combines **Threshold-Based Rules**, **VADER (Valence Aware Dictionary and sEntiment Reasoner)**, the no-text VLM fallback, and **Fuzzy Logic**. Together, these components govern live prompting, session-level scoring, and week-level DSI computation.

1. Threshold-Based Heuristics: For **live prompting**, the system first applies duration-based gates before deeper inference. Recent JITAI and infinite-scroll studies support burden-aware timing and treat sub-10-minute use as relatively brief while placing sustained scrolling in the 10 to 20+ minute range (Fiedler et al., 2024; Genugten et al., 2025; Hsu et al., 2025; Meinhardt et al., 2025; Rixen et al., 2023; Teepe et al., 2021; Terzimehić &

Aragon-Hahner, 2022; Wang et al., 2023). Accordingly, the system uses 15 minutes as the study-defined live-prompt gate and cooldown interval.

- **Session Duration Gate:** deeper prompting begins after 15 minutes of continuous use.
- **Intervention Cooldown:** after a prompt is shown, the system requires another 15 minutes of qualifying use before a new prompt can be triggered.

Mean Video Dwell Time still contributes to the fuzzy engine once prompt eligibility is met, but it is not implemented as a separate hard pre-inference gate.

Logged session summaries are retained whether or not a live prompt occurs.

2. VADER Sentiment Analysis: VADER is a lexicon- and rule-based sentiment analysis tool designed for social-media text and suited to on-device use (Hutto & Gilbert, 2014).

- **Input:** Extracted caption or comment text (e.g., “This is heartbreaking 🥺”).
- **Process:** Tokenization and lexicon lookup.
- **Output:** A Compound Score ranging from -1.0 (Most Negative) to $+1.0$ (Most Positive). Mathematically, VADER computes this bounded score via the normalization scalar:

$$C = \frac{x}{\sqrt{x^2 + \alpha}}$$

Where x is the sum of the valence scores of recognized words, and α is the standard normalization constant ($\alpha = 15$).

The implementation also retains VADER’s native UTF-8 emoji valence parsing, so emoji-dominant comments (e.g., “🥺🥺” or “💀💀💀”) still contribute affective signal even when lexical Taglish syntax is highly out-of-vocabulary.

Filipino Lexicon Extension: To handle Filipino and Taglish expressions, the system extends VADER with a limited **Minimum Viable Lexicon** (MVL). Recent low-resource sentiment work and lexicon-curation studies support this approach for code-mixed,

resource-constrained settings (Mohammed & Prasad, 2023; Nazir et al., 2026; Tho et al., 2021; Wijayanti & Arisal, 2021).

1. **Corpus Collection:** Before participant deployment, up to 50 high-frequency non-English tokens are compiled from exploratory extraction logs produced during researcher-controlled test sessions or earliest non-study dry runs. The 50-term cap is a study-defined feasibility rule that keeps the lexicon focused on dominant recurrent tokens.
2. **Expert Polarity Rating:** The collected tokens are presented to a panel of three Filipino language teachers and/or linguistics students, who independently rate each token on a sentiment polarity scale from -4 (most negative) to $+4$ (most positive), following VADER’s valence-rating convention.
3. **Inter-Rater Agreement Screening:** A token is retained only if all 3 raters assign values within ± 1 of the mean. Recent sentiment-annotation work shows that agreement can be difficult even with explicit instructions, so the ± 1 screen is used as a conservative agreement rule for retaining only tokens with comparatively stable local judgments before lexicon integration (Krušić, 2024; Äyräväinen et al., 2025).
4. **Integration:** The retained tokens and their mean valence scores are appended to the VADER lexicon dictionary at runtime, enabling the sentiment engine to recognize common Filipino expressions such as “nakakalungkot” (sad), “nakakagalit” (infuriating), or “sana all” (envious aspiration).

This approach keeps the vocabulary focused on recurring tokens while preserving annotation quality and computational efficiency for edge deployment.

3. Taglish Syntax Handling and Fallback Logic: The system uses a lightweight lexicon-based approach to keep inference on-device under mobile resource and privacy constraints. Local and low-resource studies support this choice for Philippine social-media text and

code-mixed sentiment analysis (Co et al., 2022; Cruz & Balahadia, 2022; Mohammed & Prasad, 2023; Nazir et al., 2026; Pacol & Palaoag, 2021).

To address negations such as “hindi maganda” (“not good”), the system implements a localized negation heuristic. A predefined list of common Filipino negation markers (e.g., “hindi”, “di”, “wala”, “wag”) is integrated into VADER’s pre-processing step. If a negation marker is detected within three tokens preceding a recognized sentiment token, the polarity of that token is inverted before the final compound score is computed.

Informal social-media text also contains slang, textual variation, and deliberate misspellings. Unmatched variants remain unmatched and contribute to the OOV calculation rather than being force-mapped into a sentiment label through fuzzy matching.

Dialect-heavy text and purely visual items can still fall outside the reliable text path. The system addresses that case through the no-text VLM fallback:

3.5.3. No-Text Sentiment Fallback via Moondream 0.5B

Short-form video environments can include purely visual items without accompanying text payloads. For that case, the system uses a **Per-Video Routing Heuristic** that calls a VLM fallback built on Moondream 0.5B. Moondream 0.5B is used as a small deployment-fit VLM for on-device edge inference (Lee et al., 2024; Sharshar et al., 2025).

- **Dynamic Routing Logic:** When transitioning to a new item in the feed, if the Accessibility Service successfully extracts usable text payload (caption or visible comments), the item is evaluated via VADER and routed into the text calculation tier. If no usable text is present (e.g., purely visual content without captions), the item enters the Accessibility Service screenshot plus Moondream 0.5B path. High-OOV text is not rerouted into the VLM path; it is handled through the same reliability safeguard that triggers the fallback mode.

- **Data Extraction via AccessibilityService.takeScreenshot:** Visually routed items use an on-demand screenshot after the new item has settled. This is a resource-aware compromise supported by related affective-video and privacy-compliant video analysis work showing that sparse visual evidence can support practical inference under resource constraints (Augusma et al., 2023; Qu et al., 2025; Sharma et al., 2023). To maintain the edge-computing privacy boundary, the captured frame exists only in volatile RAM for VLM inference, ensuring no user screen data is committed to persistent storage.
- **VQA Zero-Shot Computation:** The transient screenshot is processed entirely on-device by the Moondream 0.5B VLM/VQA module. The system uses constrained Visual Question Answering (VQA), and the no-text item is assigned one of five labels: SEVERE_NEG, MILD_NEG, NEUTRAL, MILD_POS, or SEVERE_POS. If the label is SEVERE_NEG or MILD_NEG, the item contributes one negative resolved unit to NSD; otherwise it contributes one resolved non-negative unit.

This route extends NSD coverage to no-text items while remaining within the device’s privacy boundary. It remains an item-level visual classification path rather than a full multimodal interpretation of viewed media (Cortiñas-Lorenzo & Lacey, 2024; Das & Singh, 2023; Johnson et al., 2025; Sharshar et al., 2025; Wei et al., 2021; Zhang & Xu, 2023).

Safety Lock: If the no-text VLM path fails because AccessibilityService.takeScreenshot is unavailable, a screenshot cannot be captured, or VLM inference does not produce a stable item label, the system drops to 2-input fuzzy inference (Video Dwell Time + Session Duration only). Sessions dominated by high-OOV text follow the same fallback path. In either case, unresolved sessions are marked **Sentiment-Unreliable** and excluded from main Doomscroll Severity Index (DSI) statis-

tical evaluation. The degraded rule base and prompt-mapping thresholds are detailed in Appendix B.

Operationally, each viewed item stays on the text path when usable text exists and moves to the VLM fallback only when no usable text exists. If the session-level OOV ratio reaches 50% or more, or if no-text resolution fails, the session is marked **Sentiment-Unreliable**: the sentiment input is dropped, inference proceeds on Video Dwell Time and Session Duration only, Level 3 prompts are disabled, and the session is excluded from main DSI computation. The 50% OOV screen functions as a study-defined reliability rule, and the study reports the number and proportion of sentiment-unreliable sessions to show DSI coverage.

4. Fuzzy Logic Inference System: The system uses fuzzy logic because the target construct is gradual, the rule base must remain inspectable for SME review, and labeled session data are not available for supervised on-device modeling (Pickering et al., 2025; Vashishtha et al., 2023). The fuzzy engine operates at the per-session inference level, while the statistical tests in Chapter 4 are used later for sample-level evaluation. The membership boundaries below define the fixed scoring framework for week-level DSI across both study weeks. In Week 2 live prompting, Session Duration and NSD memberships may be personalized from Week 1 sentiment-reliable sessions, while Video Dwell Time remains fixed (Ikegaya et al., 2025).

- **Fuzzification:** Crisp inputs (Video Dwell Time in s, Negative Sentiment Density in %, Session Duration in min) are converted into fuzzy sets using triangular membership functions. Mathematically, a triangular membership function with bounds (a, b, c) is defined as:

$$\mu_A(x) = \begin{cases} 0 & \text{if } x \leq a \\ \frac{x-a}{b-a} & \text{if } a < x < b \\ \frac{c-x}{c-b} & \text{if } b \leq x < c \\ 0 & \text{if } x \geq c \end{cases}$$

With three input variables and three linguistic states per variable (Low, Medium, High), the rule base contains $3^3 = 27$ rules. The table below presents the membership boundaries used for analytic scoring:

Table 3. Fuzzy Set Membership Boundaries for Analytic Scoring

Variable	Low	Medium	High
Video Dwell Time (s)	0–5	4–20	15–30+
Negative Sentiment Density (%)	0–33	17–83	67–100
Session Duration (min)	0–10	8–20	15–40+

These boundaries combine recent app-use timing evidence with a regular three-part partition for NSD, where the literature does not provide standard per-session negativity thresholds (Azam et al., 2021; Casalino et al., 2022; Cho et al., 2021; Ismail & Al Thani, 2022; Khairuddin et al., 2021; Meinhardt et al., 2025; Muise et al., 2024; Terzimehić & Aragon-Hahner, 2022; Tian et al., 2021). They serve as the pilot scoring structure for this estimator. Session Duration and Negative Sentiment Density act as the main risk axes, while Video Dwell Time mainly adjusts risk by showing how sustained attention is distributed across viewed items.

The table reports support intervals rather than every triangular parameter. In implementation, each Low set is a left-shoulder triangle anchored at zero, each Medium set peaks at the midpoint of its displayed support interval (12 s, 50%, and 15 min), and each High set is a right-shoulder triangle that saturates at the upper cap (30 s, 100%, and 40 min). This setup was chosen because it is the simplest piecewise-linear form that matches the declared Low/Medium/High meanings, keeps adjacent overlap without gaps, prevents overlap between non-adjacent sets, and keeps every parameter easy to check against the table.

The boundaries are interpreted as follows:

- **Video Dwell Time:** Low (0–5 s) represents immediate skip or near-skip behavior, while High (15–30+ s) represents clearly sustained attention to a single item.

- **Negative Sentiment Density:** Low, Medium, and High are defined through an evenly distributed three-term partition over the 0-100% NSD scale because the literature does not provide standard per-session negativity thresholds.
- **Session Duration:** Low (0–10 min) represents brief checking, while High (15–40+ min) represents clearly extended sessions.

The rule base is arranged so that increasing Session Duration should not lower risk when the other inputs are fixed, and increasing Negative Sentiment Density should not lower risk. Increasing Video Dwell Time generally raises risk because it reflects more sustained attention to a single item, but that effect is weaker and more context-dependent than the duration and negativity axes. The resulting rule surface is therefore mostly monotonic rather than strictly monotonic.

These ranges are retained as the fixed analytic scoring framework for week-level DSI across both study weeks and as the Week 1 live default. After the 7-day baseline window, if at least 10 sentiment-reliable sessions are available, the live prompt engine updates Session Duration and NSD memberships from that participant's Week 1 $\frac{Q_{25}}{Q_{50}}$ $\frac{Q_{75}}{Q_{95}}$ quantiles; otherwise the default memberships are retained. The ≥ 10 threshold is used to avoid unstable personalization from sparse baseline data. Video Dwell Time remains fixed in both analytic and live modes.

Sensitivity Analysis of Fuzzy Boundaries: Post-deployment, the study conducts a one-at-a-time local sensitivity analysis of the nine boundary parameters by varying each parameter across a pre-specified plausibility range while the others are held constant. For each perturbation, the analysis reports: (1) mean absolute score change, (2) percentage of sessions reclassified (Safe/Warning/Critical), and (3) which parameters produce the largest output instability under the tested range (Dogan, 2021; Shahari & Rasmani, 2024; Shukla & Muhuri, 2025; Vinogradova-Zinkevič, 2023).

- **Rule Evaluation:** The system evaluates a rule base of 27 rules (3 variables $\times$ 3 levels each = 3^3 combinations). **Safe** is assigned to clearly low-risk states and isolated mild signals. **Warning** is assigned to mixed or moderate states without strong multi-axis convergence. **Critical** is assigned when prolonged use and negative exposure converge, with Video Dwell Time acting mainly as an intensifier. The rule assignments follow these principles:

- every Low/Medium/High combination is assigned exactly once, so the engine has no undefined corner cases
- no rule is **Critical** when Session Duration and Negative Sentiment Density are both Low
- prolonged duration alone does not produce **Critical** when Negative Sentiment Density remains Low
- High NSD is escalated most strongly when at least one other axis is also non-low, especially Session Duration
- High Dwell Time alone is insufficient for **Critical** and mainly intensifies already concerning states
- **Warning** remains the default mixed class whenever the evidence is concerning but not yet strongly convergent

Read in groups, Rules 1, 2, 4, 10, and 19 are assigned **Safe** because at least two axes remain low. Rules 3, 5-8, 11-14, 16, and 20-22 are assigned **Warning** because they show one elevated axis or mixed evidence without strong duration-negativity convergence. Rules 9, 15, 17-18, and 23-27 are assigned **Critical** because prolonged use and negative exposure converge, or because High NSD is reinforced by High Dwell Time even when the total session is still short.

Rule 15 (Medium Dwell, Medium NSD, High Duration) is classified as **Critical** because the session is already prolonged and both other axes are non-low. Rule 23 (High Dwell,

Medium NSD, Medium Duration) is also classified as **Critical** because sustained attention and non-low negative exposure already converge in a session that is no longer brief. Rule 25 (High Dwell, High NSD, Low Duration) reflects the same logic in concentrated form: even a short session can show concentrated negative immersion when viewed items are mostly negative and each item holds attention for a relatively long time.

Representative rules include:

Table 4. Representative Appendix A Rules (6 of 27)

No.	Video Dwell Time	Negative Sentiment Density	Session Duration	Risk Level
1	Low	Low	Low	Safe
14	Medium	Medium	Medium	Warning
27	High	High	High	Critical
26	High	High	Medium	Critical
9	Low	High	High	Critical
19	High	Low	Low	Safe

The complete 27-rule inference base is presented in Appendix A.

Exception Logic: Most rules are arranged so that higher Session Duration or higher Negative Sentiment Density does not reduce risk, but one deliberate exception is retained for interpretive accuracy. Appendix A Rule 9 (Low Dwell, High NSD, High Session Duration) is classified as **Critical** because low dwell should not automatically downgrade a session that is both prolonged and dominated by negative items; in that edge case, rapid chaining across many negative items is treated as more concerning than reassuring. By contrast, Appendix A Rule 19 (High Dwell, Low NSD, Low Session Duration) is not treated as an exception but as a boundary clarification: sustained attention to a single item, without prolonged session time or negative-dominant exposure, is not enough by itself to indicate doomscrolling. This is why the table treats dwell time as a modifier, not the main driver.

- **Defuzzification:** The aggregated fuzzy output is converted back into a crisp “Risk Score” (0–100) using the Center of Gravity (CoG) method. The three output classes occupy equal thirds of the reporting scale, with category cutoffs at 33.33 and 66.67 and output centers at 16.67, 50.00, and 83.33 for Safe, Warning, and

Critical, respectively. Formally, this computation involves rule activation weight (w_i) following the MIN T-norm, and output defuzzification:

$$w_i = \min(\mu_{\text{Dwell}}(x), \mu_{\text{NSD}}(y), \mu_{\text{Duration}}(z))$$

$$\text{Risk Score} = \frac{\sum_{i=1}^n w_i \cdot c_i}{\sum_{i=1}^n w_i}$$

Where w_i is the activation weight of rule i , and c_i is the crisp output center for the mapped risk class. For each fired rule, the smallest membership value determines the rule's activation weight. This prevents one strong input from fully dominating when the other conditions required by that rule remain weak.

Dynamic Weighting vs. Static Percentages: The adopted MIN-based fuzzy inference does not assign flat percentage weights to individual variables. Each fired rule is limited by its weakest required antecedent, so a variable's influence depends on the full combination of inputs rather than on a fixed linear contribution.

Illustrative Worked Example: Given the input tuple (VideoDwellTime = 45 s, NegativeSentimentDensity = 62%, SessionDuration = 35 min), the system computes Video Dwell Time High membership = 1.0, Negative Sentiment Density Medium membership ≈ 0.64 , and Session Duration High membership = 0.80. The dominant firing pattern is Appendix A Rule 24 (High Dwell Time, Medium NSD, High Duration $\rightarrow$ Critical), so the defuzzified Risk Score is 83.33/100 and the resulting intervention level is Level 3 (Critical).

3.6. Method in Evaluation of the Study

The study uses four complementary evaluation methods: the **ISO/IEC 25010 Software Quality Model** for technical quality assessment, TAM for user acceptance evaluation, an SME panel for expert review, and a baseline-intervention comparison of selected logged metrics and self-reported doomscrolling scores. Results are reported separately as **operational feasibility** (successful deployment completion and absence of critical system

failures), **software acceptability** (ISO/IEC 25010, TAM, and SME results), and **estimator plausibility** (baseline convergent association together with sentiment-reliable coverage).

3.6.1. ISO/IEC 25010 Software Quality Evaluation

The study specifically focuses on four key characteristics of the ISO/IEC 25010 model (International Organization for Standardization, 2023) relevant to privacy-sensitive mobile well-being applications:

1. **Functional Suitability:** Assessing whether the heuristic risk-state estimation and intervention features function as intended and cover the specified requirements.
2. **Performance Efficiency:** Evaluating the application's resource usage through both user survey responses and objective profiling metrics such as CPU usage, RAM consumption, and battery drain observed during controlled testing on a small reference-device set, with field-deployment logs used only as descriptive context.
3. **Usability:** Measuring the ease of use, learnability, and user satisfaction with the interface and intervention mechanisms.
4. **Reliability:** Assessing the system's stability and availability during extended periods of monitoring without crashing or being killed by the Android OS.

3.6.2. Technology Acceptance Model (TAM) Evaluation

To assess user acceptance, the study uses the Technology Acceptance Model (TAM). Recent mHealth acceptability reviews continue to treat TAM as a practical user-acceptance framework, with Perceived Usefulness and Perceived Ease of Use retained as core constructs (Adnan et al., 2025). Two TAM constructs are measured:

1. **Perceived Usefulness (PU):** The degree to which users believe the system enhances their ability to manage doomscrolling behavior.

2. **Perceived Ease of Use (PEOU):** The degree to which users believe the system is free of effort to learn and operate.

Each construct is measured using 6 Likert-scale items aligned to these two core TAM constructs and adapted to the doomscrolling risk-estimation context. The TAM instrument is administered in the Phase 5 post-usage survey.

3.6.3. Subject Matter Expert (SME) Evaluation

A panel of **three (3) subject matter experts** evaluates the system's technical quality and risk-estimation logic. The SME panel is composed of:

- One (1) expert in software engineering or mobile application development
- One (1) expert in data science, machine learning, or fuzzy logic systems
- One (1) expert in digital well-being, behavioral psychology, or educational technology

Each SME evaluates the system using a structured rubric covering: (1) the technical soundness of the hybrid risk-estimation framework, including VADER integration and fuzzy inference behavior, (2) the plausibility of the selected input ranges for Video Dwell Time, Negative Sentiment Density, and Session Duration, (3) the internal coherence of the 27-rule base, including whether nearby rules change in a sensible direction across duration and negative exposure and whether exception rules such as rapid negative-content chaining are reasonable, (4) the quality of the system architecture and privacy implementation, (5) the appropriateness of the intervention design, and (6) overall software quality using the same ISO/IEC 25010 characteristics evaluated by end users. SME responses are collected using a 5-point Likert scale and analyzed descriptively (mean, standard deviation) alongside qualitative feedback. This SME review serves as narrative expert appraisal of the rule logic and as an expert plausibility check, not as formal empirical calibration or a content-validity index.

3.6.4. Testing Method

The study uses both **black box** and **white box** testing. Black-box testing checks whether visible functions behave as expected under realistic use, while white-box testing checks whether the internal scoring, routing, and data-flow logic behave correctly at code and module level.

3.6.4.1. Black Box Testing Procedure

The black-box procedure evaluates the application through expected inputs, user actions, and observed outputs without relying on internal code inspection. The full application is deployed on a small set of physical devices representing the screen sizes and Android versions available to the researchers to verify that usage patterns (e.g., prolonged session with high negative content) produce the expected risk classification and intervention, including correct routing of no-text cases into the VLM fallback and correct degradation to the 2-input fallback when sentiment remains unresolved. The same procedure checks user-visible flows such as onboarding, permission handling, dashboard display, logging continuity, and prompt presentation.

3.6.4.2. White Box Testing Procedure

The white-box procedure evaluates the internal logic and data flow of the system. The Kotlin-implemented VADER-compatible scoring engine and fuzzy-logic rules are tested in isolation using JUnit to verify that known inputs produce expected outputs. The Accessibility Service-to-database pipeline is also tested to confirm correct data flow and that interventions trigger from logged state changes. This same layer verifies text-path routing, no-text VLM routing, and expected fallback to the 2-input safety mode when sentiment remains unresolved.

3.7. Data Collection Scheduling

Data collection follows five sequential phases. Phase 1 covers recruitment, screening, consent, and group assignment. Phase 2 covers baseline profile completion and application setup. Phase 3 covers the one-week baseline logging period with prompts disabled. Phase 4 covers the one-week deployment period in which only the intervention group receives adaptive prompts while the control group continues logging-only. Phase 5 covers the post-usage survey and final retrieval of aggregate local logs.

3.8. Data Gathering

Data gathering is conducted through five phases:

Phase 1: Recruitment, Screening, and Consent Potential participants are recruited through non-probability purposive-convenience sampling using accessible channels such as academic networks, peer referrals, and online groups. Interested individuals undergo screening to confirm eligibility based on age, active use of at least one target short-form video platform, and study-device compatibility, including whether the phone can install the study build, maintain the required permissions, and keep the monitoring service active under ordinary use. Eligible participants are then provided with an informed consent form explaining the study's purpose, procedures, data to be collected, risks, safeguards, and right to withdraw. After consent and eligibility confirmation, participants are assigned to either the intervention group or the logging-only control group using the concealed permuted-block allocation list described earlier, with an initial 1:1 allocation target of 25 per group. Both groups receive the same baseline setup and Week 1 logging procedure; the distinction becomes operational only in Week 2 when prompts are enabled for the intervention group.

Phase 2: Baseline Profile Form and Application Setup Prior to deployment, each participant completes a baseline profile form capturing demographic information, estimated short-form video use, and general platform-use habits. The participant is then assisted in installing the application and enabling the necessary Android permissions. Each participant

is assigned a unique study code used to link logs and survey forms; the name-to-code list is retained separately by the researchers. This form provides contextual information but is not treated as the primary source of behavioral measurement.

Week 1 provides the fixed-prior baseline DSI and self-report anchor. At the end of the 7-day baseline, the live prompt engine may lock participant-specific Session Duration and NSD memberships from Week 1 sentiment-reliable sessions while leaving Video Dwell Time fixed. This follows recent JITAI evidence favoring personalized intervention criteria over uniform ones for live prompting (Ikegaya et al., 2025). If reliable baseline coverage is insufficient, the default priors are retained.

Phase 3: Week 1 Baseline Logging Phase During Week 1, the app’s built-in logger records quantitative usage metrics locally on the participant’s device while adaptive intervention prompts remain inactive. This baseline period establishes the participant’s observed usage pattern under the study conditions and provides the baseline DSI and self-report anchor used for convergent association testing.

The Week 1 logged data include:

- **Usage Metrics:** Session duration, video dwell time, and supporting interaction counts such as swipe totals.
- **Sentiment-Related Indicators:** Session-level sentiment scores and Negative Sentiment Density estimates.
- **Reliability Events:** Application crashes, service interruptions, and related system-stability notes.
- **Performance Indicators:** CPU, RAM, and battery observations gathered through limited in-app logging and controlled profiling on a small reference-device set, with heterogeneous participant-device traces used descriptively.

At the conclusion of Week 1, participants complete the short-form 4-item **Doomscrolling Scale** (Satici et al., 2023; Sharma et al., 2022), using Items 1, 2, 10, and 12 of the original

instrument and a 7-day recall instruction stem (e.g., “In the past week...”; see Appendix D), to provide the baseline self-report anchor for convergent association. The original instrument was validated as a general self-report scale, so changes between administrations are interpreted as short-horizon self-report differences.

Phase 4: Week 2 Deployment Phase During Week 2, **intervention-group** participants receive adaptive digital mindfulness prompts through a live prompt engine that retains the fixed rule base and fixed Video Dwell Time memberships but, when available, replaces the global Session Duration and NSD memberships with participant-specific Week 1 quantile-derived bounds. **Control-group** participants continue logging-only operation with prompts disabled via a configuration flag. Both groups continue to have usage and sentiment-related metrics logged, while prompt-response data for the intervention group are recorded descriptively.

For behavioral comparison, raw elapsed session duration and raw elapsed video dwell time are treated as the primary time-based metrics. **Prompt-excluded active-use metrics** are also retained: when a prompt appears, the active content-view interval is closed, and time spent inside the system-generated intervention is excluded until the participant returns to the target platform. These prompt-excluded metrics are reported as supplementary traces of prompt-interrupted use. Observed Week 2 differences are interpreted as short-term behavioral differences under the study conditions.

Potential confounds include time-of-day and day-of-week variation, external events during the two-week window, and device-specific differences in processing speed or background-process management. The randomized two-group design helps balance these factors across study arms, while the within-subject Week 1 to Week 2 comparison in the intervention group accounts for stable individual differences.

At the conclusion of Week 2, participants complete the same short-form 4-item **Doomscrolling Scale** under the same 7-day recall instruction stem (Appendix D) to provide the post-intervention self-report anchor for paired comparison.

Phase 5: Post-Usage Survey and Final Data Retrieval To align data structures for baseline convergent association and descriptive week-level comparison, logged session summaries over each 7-day period are aggregated into a composite **Doomscroll Severity Index** per user per week. This analytic DSI uses the same fixed-prior scoring framework across both study weeks. Unlike the live Week 2 prompt engine, the DSI engine is not personalized after Week 1, so week-level summaries remain on one fixed scale:

$$DSI_w = \left(\frac{1}{m_w} \right) \sum_{s=1}^{m_w} RiskScore_s$$

where m_w is the number of week- w sessions classified as sentiment-reliable and therefore doomscrolling-evaluable. Sessions flagged as **Sentiment-Unreliable** are excluded from DSI computation. DSI is a mean of retained session scores rather than a cumulative sum, so these exclusions affect coverage more than scale. Week-specific DSI values are retained for descriptive comparison, convergent testing is anchored to the baseline week only, and DSI coverage is reported by week and study arm in terms of retained participants, retained sessions, and excluded sentiment-unreliable sessions.

Interpretation of Classification Metrics: Traditional supervised classification metrics (e.g., precision, recall, F1, AUROC) are not estimated because session-level ground-truth labeling would require interrupting active scrolling. The study therefore uses convergent association as the main plausibility check and interprets the system as a heuristic behavioral risk estimator.

After the deployment period, participants complete a structured post-usage survey instrument with three components: (a) the ISO/IEC 25010 software quality evaluation, (b) the TAM assessment, and (c) open-ended feedback items. At the same stage, aggregate local

logs are exported using the participant study code only and transferred to a password-protected research folder accessible only to the researchers.

For the ISO/IEC 25010 component, the survey uses a 5-point Likert Scale to evaluate the four target quality characteristics: Functional Suitability, Performance Efficiency, Usability, and Reliability. For the Usability sub-characteristic, all 10 items of the System Usability Scale (SUS) are adopted verbatim, and the standard SUS scoring formula is applied to yield a composite usability score on a 0–100 scale. The SUS is a standardized usability measurement tool whose validity for digital health applications has been supported by a meta-analysis of 114 apps yielding a benchmark mean of 68.05 (SD 14.05) (Hyzy et al., 2022). For the remaining three sub-characteristics (Functional Suitability, Performance Efficiency, and Reliability), the study uses researcher-developed items derived from the ISO/IEC 25010 sub-characteristic definitions (International Organization for Standardization, 2023) and refined through pre-deployment expert review.

For the TAM component, 12 Likert-scale items (6 for Perceived Usefulness, 6 for Perceived Ease of Use) aligned to the study’s contextualized TAM constructs are included to assess user acceptance of the system.

3.8.1. Instrument Review and Reliability Safeguards

Before deployment, the combined survey instrument undergoes structured expert content review to check item relevance, clarity, redundancy, and alignment with the study constructs. Reviewer comments are consolidated, and unclear or overlapping items are revised before administration.

After data collection, internal consistency for each multi-item sub-scale is estimated through Cronbach’s alpha and interpreted as an index of item coherence, with appropriate caution given the pilot sample size and the mix of standardized and adapted items.

Open-ended feedback items are used as descriptive support for the survey findings. Two researchers independently review the responses, reconcile coding differences through discussion, and report recurring issue categories.

The alignment between the study's research variables, data sources, and instruments is summarized in the following table.

Table 5. Variable–Instrument Alignment Matrix

Research Variable	Data Source	Instrument / Tool	Phase
Baseline usage profile	Self-report / Logs	Baseline profile form + Week 1 logs	Phase 2 & 3
Session Duration	Automated system logs	Built-in application logger	Phase 3 & 4
Video Dwell Time (s)	Automated system logs	Built-in application logger	Phase 3 & 4
Negative Sentiment Density (%)	Automated system logs	VADER text-path + VLM no-text path + Built-in logger	Phase 3 & 4
Prompt-response profile (descriptive)	Automated system logs	Built-in application logger	Phase 4
Pre/Post Doomscrolling Scale Score	Self-report	Short-form 4-item Doomscrolling Scale (Appendix D)	End of Phase 3 & 4
Baseline convergent association	Week 1 DSI + Week 1 Doomscrolling Scale	Bivariate Correlation (Pearson's r / Spearman's ρ)	End of Phase 3
Functional Suitability	Likert-scale survey	Researcher-developed ISO/IEC 25010 items + expert review	Phase 5
Performance Efficiency (subjective)	Likert-scale survey	Researcher-developed ISO/IEC 25010 items + expert review	Phase 5
Performance Efficiency support (objective profiling)	Profiling records / logs	Controlled profiling + field-deployment metrics	Phase 3 & 4

Table 5. Variable–Instrument Alignment Matrix (continued)

Research Variable	Data Source	Instrument / Tool	Phase
Usability	Likert-scale survey	System Usability Scale (SUS), benchmarked in Hyzy et al. (2022)	Phase 5
Reliability	Likert-scale survey	Researcher-developed ISO/IEC 25010 items + expert review	Phase 5
Perceived Usefulness (TAM)	Likert-scale survey	Contextualized TAM PU items	Phase 5
Perceived Ease of Use (TAM)	Likert-scale survey	Contextualized TAM PEOU items	Phase 5
Expert quality assessment	SME evaluation	Structured rubric (ISO 25010 + risk-estimation logic)	SME

3.9. Respondents of the Study and Sampling Technique

The respondents of this study are Filipino Android users aged 18 years and above who actively use at least one of the target short-form video platforms. The findings apply to this target population and to the non-probability nature of the sample. The sampling technique employed is **purposive-convenience sampling** because participants must meet specific technical and behavioral eligibility criteria and be reachable through channels accessible to the researchers.

The study reports how many participants used TikTok, Facebook Reels, and Instagram Reels during the field evaluation, including platform mix by study arm. Per-platform extraction success rates and analyzable-session counts are also reported, and cross-platform interpretation is limited to platforms that yielded stable extraction throughout the deployment window.

Inclusion Criteria:

- Must be aged 18 years or older to provide independent informed consent.
- Must own an Android smartphone (Version 8.0 or higher) that passes study-compatibility screening for installation, permissions, and background-service stability.
- Must be an active user of at least one target short-form video platform (TikTok, Facebook Reels, or Instagram Reels).
- Must consent to the installation of the application and the collection of usage data.

Sample Size Justification: A minimum of $N = 50$ participants is targeted, split as evenly as possible into an intervention group and a control group, with an initial allocation target of $n = 25$ per group. This target supports a balanced pilot deployment and post-usage evaluation within an undergraduate thesis scope; it is not intended as a fully powered efficacy sample. Analyses that exclude sentiment-unreliable sessions or participants with insufficient baseline data report their effective sample sizes separately. With a minimum sample of about

50 participants, between-group estimates are interpreted with emphasis on effect sizes and 95% confidence intervals alongside p-values.

Selection Bias Acknowledgment: Purposive-convenience sampling may overrepresent users who are already receptive to self-regulation tools. This self-selection tendency can affect perceived acceptance and compliance estimates.

3.10. Statistical Treatment

Research Question 1 is addressed descriptively through the design-and-development sections earlier in this chapter and is not treated as an inferential hypothesis test. The evaluation data are analyzed using the following statistical methods, each selected for a specific data source and objective.

1. Descriptive Summaries (Survey Data — ISO 25010 & TAM): The Likert-scale responses from the ISO/IEC 25010 software quality evaluation survey and the TAM instrument (Perceived Usefulness and Perceived Ease of Use subscales) are summarized using **mean scores** and standard deviation.

Mean scores are described on a 5-point Likert scale, where 3.00 represents the neutral midpoint. For evaluative decisions, the study uses pre-set favorable thresholds of ≥ 3.50 for ISO/IEC 25010 and TAM constructs and ≥ 4.00 for SME ratings. These are study-defined favorable benchmarks rather than universal cutoffs.

2. Objective Acceptance Thresholds for Evaluation: To reduce post-hoc rationalization of evaluation results, the following pre-set interpretation thresholds are established as decision aids for Chapter 4 rather than as stand-alone proof of software success:

- **System Usability Scale (SUS):** A composite score of ≥ 70 is treated as the study's usability target, set slightly above the 68.05 digital-health benchmark mean reported by Hyzy et al. (2022).

- **ISO/IEC 25010 Subscales:** A mean score of ≥ 3.50 out of 5 is treated as the study-defined favorable-evaluation target for Functional Suitability, Performance Efficiency, and Reliability.
- **TAM Constructs:** A mean score of ≥ 3.50 is treated as the study-defined favorable-evaluation target for Perceived Usefulness and Perceived Ease of Use.
- **SME Evaluation:** A mean rating of ≥ 4.00 out of 5 is treated as the study-defined favorable-evaluation target for expert review.

Objective CPU, RAM, and battery observations are summarized descriptively across the deployment and controlled profiling sessions. Given device differences in hardware and manufacturer background-process behavior, these profiling results are interpreted as supporting evidence for Performance Efficiency instead of as a uniform decision criterion. Results below these thresholds are reported as unmet software-acceptability targets in Chapter 4. Operational feasibility and estimator plausibility are reported separately. Given the pilot-sized sample and the possibility of analyzable-sample reduction from sentiment-unreliable sessions, inferential results are interpreted primarily through effect sizes, confidence intervals, and coverage rather than pass-fail significance alone.

3. Cronbach's Alpha (Instrument Reliability): To assess internal consistency, **Cronbach's alpha** (α) is computed for each sub-scale (Functional Suitability, Performance Efficiency, Usability, Reliability, PU, PEOU). A pre-set coefficient threshold of $\alpha \geq 0.70$ is adopted as a conventional descriptive benchmark for this study. Given the pilot sample and the mix of standardized and adapted items, alpha is treated as a reliability check rather than as a stand-alone validation claim.

4. Primary Behavioral Analysis: The main quantitative comparison for Research Question 2 uses an **Independent Samples t-test on change scores** (Week 2 minus Week 1) to compare observed short-term behavioral differences between the intervention group and the control group. Using change scores rather than raw Week 2 values adjusts for chance

baseline imbalance and individual differences. As a secondary within-subject analysis, a **Paired Samples t-test** compares Week 1 and Week 2 within the intervention group. Given the pilot sample, these tests are treated as exploratory analyses that estimate short-term observed differences under the study conditions rather than precise intervention-effect magnitudes. Separate primary tests are conducted for:

- Mean daily **raw elapsed** session duration (minutes)
- Mean daily **raw elapsed** video dwell time (s)
- Mean daily Negative Sentiment Density (%)
- Pre/Post Doomscrolling Scale Score (Self-Reported)

For the Negative Sentiment Density outcome, mean daily NSD is computed only from sentiment-reliable sessions. Days or participant-weeks with no sentiment-reliable sessions contribute missing data for that specific outcome and are excluded only from that specific NSD comparison, with the analyzed n reported.

As a supplementary transparency check, the same tests are also conducted on **prompt-excluded active-use metrics** for session duration and video dwell time. The difference between raw elapsed and prompt-excluded results is reported to show the measurement-rule effect.

To control family-wise Type I error across the four change-score comparisons, p-values are adjusted using the **Holm-Bonferroni sequential correction** at an initial family-wise $\alpha = 0.05$. For each comparison, **Cohen's d** and its 95% confidence interval are reported alongside the Holm-adjusted p-value to support interpretation of effect size, not statistical significance alone.

Handling of Assumption Violations: For the primary independent-samples t-test, normality of change scores within each group is checked with Shapiro-Wilk and equality of variances with Levene's test. If normality is severely violated, the **Mann-Whitney U test** is

used as fallback. If variances are unequal but normality holds, Welch's t-test is applied. For the secondary paired t-test, normality of within-subject differences is checked with Shapiro-Wilk; if violated, the **Wilcoxon Signed-Rank Test** is used. If analyzable sample sizes become too small for stable inferential interpretation after outcome-specific exclusions, the affected comparison is reported descriptively with its effective n and directional pattern rather than being overstated as confirmatory evidence.

5. Descriptive Analysis of Prompt-Response Logs: Week 2 prompt-response data (e.g., continue, dismiss, take-break) are summarized using frequency counts and percentages. These logs are treated as descriptive evidence of intervention interaction only and are not analyzed as a baseline-comparable outcome variable.

6. Baseline Convergent Association Analysis: Baseline convergent association is assessed on **Week 1 data only**. First, a component-wise correlation table reports the bivariate association between each individual input variable (mean daily session duration, mean daily video dwell time, mean daily NSD) and the baseline Doomscrolling Scale score. Second, the bivariate correlation between the full composite DSI (computed with fixed priors) and the Doomscrolling Scale is computed. The pattern of component-wise versus composite correlations is discussed descriptively to assess whether the multi-variable composite associates more strongly with self-reported doomscrolling than session duration alone. Only participants with at least three sentiment-reliable sessions in Week 1 are included; the effective sample size ($N_{\text{effective}}$) is reported. This coverage rule follows recent digital-measure practice for forming week-level aggregates from repeated observations, with three observations commonly used as a pragmatic weekly minimum while stronger reliability claims require more data (Buekers et al., 2025; Meyer et al., 2022; Ratitch et al., 2023; Yao et al., 2021). Here, the ≥ 3 rule functions as a pragmatic floor for the baseline aggregate. **Pearson's r** is used when linearity and normality assumptions hold; otherwise **Spearman's**

ρ . Results are interpreted as a baseline plausibility check rather than as diagnostic validation.

7. Interpretation: For these exploratory pilot analyses, the nominal level of significance is set at $\alpha = 0.05$. Statistically significant differences between Week 1 and Week 2 are interpreted as observed short-term behavioral differences under the study conditions. The two-group randomized design supports stronger inference than a one-group pre/post comparison, but findings are not interpreted as long-term intervention effects or calibrated validation. Prompt-excluded active-use metrics are interpreted separately because they incorporate the prompt-display rule.

8. Descriptive Coding of Open-Ended Feedback (Phase 5): Open-ended responses from the post-usage survey are analyzed through a light inductive coding procedure. Two researchers read the responses in full, group them into recurring issue categories, and reconcile coding differences through discussion before the categories are summarized. The goal is to identify usability issues, perceived intervention value, and implementation constraints not captured by the Likert-scale measures. This step supports interpretation of the survey results and surfaces recurring concerns for Chapter 4 discussion.

9. Missing Data and Exclusion Criteria: For the convergent correlation analysis, participants need at least three sentiment-reliable sessions in Week 1 ($m_1 \geq 3$) for Week 1 DSI to enter the correlation dataset. This rule includes the edge case $m_1 = 0$, where DSI is undefined by construction. The effective sample size ($N_{\text{effective}}$) after this exclusion is reported explicitly. For the NSD outcome, participants require at least one sentiment-reliable day in each compared week to contribute a valid week-level comparison; otherwise that specific outcome is treated as missing for that participant. For paired pre/post analyses, complete paired observations are required per outcome variable, and the analyzed n for each test is reported.

3.11. Ethical Considerations

This study involves the monitoring of human participants' social media usage behavior, and as such, strict ethical protocols are observed to protect respondent welfare, privacy, and autonomy.

Institutional Review: Prior to data collection, the research protocol, instruments, and informed consent forms are submitted for review and approval by the university's Research Ethics Committee (or Institutional Review Board). No data collection activities commence until formal ethical clearance is obtained.

Informed Consent: All participants are provided with a detailed informed consent form explaining the study's purpose, procedures, data collected, risks, and benefits. Participation is entirely voluntary. The study version requires explicit user activation of the Android Accessibility Service through the device's system settings, providing the study's operational consent layer for text, interaction access, and the service-declared screenshot capability used for no-text visual processing.

Data Protection and Privacy: Consistent with the Privacy by Design principle described in the System Architecture section, the study version processes scraped textual content and temporary no-text screen frames in RAM only, with no persistent storage of raw text data or captured frames. Incidental third-party text, such as usernames or interface labels, is processed briefly and filtered where identifiable so non-participant data is not retained. Only aggregated metrics (e.g., sentiment scores, session duration) are retained. Logs and survey instruments are linked only through a participant study code, while the name-to-code key is stored separately by the researchers. Aggregate exported logs are transferred to a password-protected research folder accessible only to the researchers. The system does not transmit any data to external servers during routine operation.

Right to Withdraw: Participants may withdraw from the study at any time without penalty. Upon withdrawal, all locally stored usage data is deleted from the participant's device. If withdrawal occurs before final dataset consolidation, any exported participant record linked to that study code is also deleted from the research dataset.

Minimization of Harm: The study does not involve clinical diagnosis or treatment of mental health conditions. The mindfulness interventions (breathing exercises, break reminders) are designed as general wellness prompts, not therapeutic interventions. The methodological and technical limitations arising from the research design and system architecture of this study are documented in Chapter 1 under **Limitations**.

Chapter 4

PRESENTATION, ANALYSIS AND INTERPRETATION OF DATA

This chapter presents the results, analysis, and interpretation of the data collected for the study. The presentation follows the statement of the problem and separates the findings into five parts: the system architecture and estimation framework, the Week 1 to Week 2 behavioral comparison, the baseline convergent association between the Doomscroll Severity Index (DSI) and self-reported doomscrolling, the user evaluation using ISO/IEC 25010 and the Technology Acceptance Model (TAM), and the subject matter expert evaluation.

All numerical entries in this draft are placeholders until the final field deployment, post-usage survey, expert evaluation, and statistical analysis are completed. No favorable or unfavorable finding is inferred in this chapter until the corresponding data are available.

4.1. Presentation of Results

The results are organized according to the five research questions. For data-based sections, the tables use the planned metrics from Chapter 3 and retain explicit placeholders for sample size, descriptive statistics, reliability coefficients, effect sizes, confidence intervals, and p-values. The interpretations following each table should be completed only after the final dataset has been cleaned and analyzed.

4.1.1. Research Question 1: Privacy-Preserving Mobile Architecture and Estimation Framework

The first research question asks what privacy-preserving mobile architecture and estimation framework can support doomscrolling-related risk estimation on the target short-form video platforms using behavioral indicators and sentiment-related indicators when reliably resolvable, with 2-input behavioral fallback for sentiment-unreliable sessions. This question is addressed through the completed system design, implementation evidence, and pre-deployment testing results rather than through participant outcome statistics alone.

Table 6. System Architecture and Estimation Framework Result Matrix

System Component	Intended Function	Implementation Evidence	Result Status
Data Access Layer	Collect target-app interaction events, visible text, app-state changes, aggregate usage signals, and no-text screenshot frames through Android Accessibility Service.	[TBD after evaluation data: insert implemented modules and test evidence]	[TBD after evaluation data: complete / partial / not met]
Business Logic Layer	Route analyzable text through the VADER-compatible path, route no-text items through the on-device VLM path, compute NSD, and apply fuzzy inference.	[TBD after evaluation data: insert implemented modules and test evidence]	[TBD after evaluation data: complete / partial / not met]
2-Input Safety Fallback	Use Video Dwell Time and Session Duration when sentiment remains unresolved or unreliable.	[TBD after evaluation data: insert fallback test result]	[TBD after evaluation data: complete / partial / not met]
Presentation Layer	Deliver awareness notification, pause prompt, breathing overlay, and dashboard feedback according to risk state.	[TBD after evaluation data: insert UI and prompt test result]	[TBD after evaluation data: complete / partial / not met]
Privacy Safeguards	Keep raw text and transient screen frames out of persistent storage and retain only aggregate metrics linked to study codes.	[TBD after evaluation data: insert privacy verification result]	[TBD after evaluation data: complete / partial / not met]

As shown in the table, the system is evaluated according to whether each planned architectural component was implemented and verified. The final interpretation should explain

which components were completed, which components required fallback behavior, and whether the implemented system remained aligned with the privacy-preserving design described in Chapter 3. At this draft stage, the table is retained as an interpretation shell and does not yet establish that the system met the objective.

Table 7. Black Box and White Box Testing Summary

Testing Area	Expected Behavior	Result	Remarks
Onboarding and Permissions	The app installs, opens, and guides the participant through required permissions.	[TBD after evaluation data: pass/fail]	[TBD after evaluation data: insert remarks]
Accessibility Extraction	The service detects target app sessions, content transitions, and visible text where available.	[TBD after evaluation data: pass/fail]	[TBD after evaluation data: insert remarks]
Text Sentiment Path	The VADER-compatible path scores analyzable text and contributes to NSD.	[TBD after evaluation data: pass/fail]	[TBD after evaluation data: insert remarks]
No-Text VLM Path	No-text items are routed to the on-device VLM fallback when Accessibility Service screenshot support and device conditions allow.	[TBD after evaluation data: pass/fail]	[TBD after evaluation data: insert remarks]
Fuzzy Inference	Known input combinations produce expected Safe, Warning, or Critical risk states.	[TBD after evaluation data: pass/fail]	[TBD after evaluation data: insert remarks]
Prompt Triggering	The appropriate intervention prompt appears when risk thresholds are reached.	[TBD after evaluation data: pass/fail]	[TBD after evaluation data: insert remarks]
Local Logging	Aggregate metrics are stored under study codes without retaining raw text or captured frames.	[TBD after evaluation data: pass/fail]	[TBD after evaluation data: insert remarks]

The testing summary should be interpreted as evidence for Research Question 1 only after final implementation testing is completed. Any failures should be described as design constraints, reliability events, or unresolved implementation limitations rather than hidden from the discussion.

4.1.2. Evaluation Method, Respondents, and Instrument

The evaluation method, respondent participation, and instruments are presented here as a completion shell. This section should be finalized only after the deployment records, post-usage survey responses, and expert evaluation forms are complete.

Table 8. Evaluation Method, Respondents, and Instrument Placeholder

Evaluation Component	Planned Source	Data to Insert	Status
Behavioral log evaluation	Week 1 and Week 2 app-generated usage records	[TBD after evaluation data: insert cleaned log availability, inclusion basis, and analysis status]	[TBD after evaluation data: complete after log validation]
User evaluation	Post-usage ISO/IEC 25010, SUS, and TAM survey instrument	[TBD after evaluation data: insert respondent count, response completeness, and instrument coverage]	[TBD after evaluation data: complete after survey encoding]
Subject matter expert evaluation	Expert evaluation form covering technical design, privacy safeguards, heuristic logic, and intervention structure	[TBD after evaluation data: insert expert count, rating completeness, and feedback coverage]	[TBD after evaluation data: complete after SME evaluation]
Open-ended feedback	Post-usage comments and expert comments	[TBD after evaluation data: insert coding status and theme count]	[TBD after evaluation data: complete after qualitative coding]

The final manuscript should use this section to identify the actual respondents, instruments, and evaluative evidence used in Chapter 4. No respondent count, survey result, or expert result is interpreted until the corresponding dataset has been verified.

4.1.3. Likert Scale and Scoring Method Placeholder

The scoring method for user and expert evaluation should be stated before the per-category results are interpreted. This section should identify the final scale anchors, interpretation ranges, and scoring procedure used for ISO/IEC 25010, TAM, SUS, and SME evaluation instruments.

Table 9. Scoring Method and Interpretation Placeholder

Instrument / Measure	Scoring Basis	Interpretation Range to Insert	Status
ISO/IEC 25010 user survey	[TBD after evaluation data: insert final Likert scale anchors and computation procedure]	[TBD after evaluation data: insert interpretation ranges after instrument finalization]	[TBD after evaluation data: complete after survey instrument finalization]
Technology Acceptance Model survey	[TBD after evaluation data: insert final Likert scale anchors and computation procedure]	[TBD after evaluation data: insert interpretation ranges after instrument finalization]	[TBD after evaluation data: complete after survey instrument finalization]
System Usability Scale	[TBD after evaluation data: insert SUS scoring procedure]	[TBD after evaluation data: insert interpretation threshold used by the study]	[TBD after evaluation data: complete after scoring procedure verification]
Subject matter expert evaluation	[TBD after evaluation data: insert expert-rating scale anchors and computation procedure]	[TBD after evaluation data: insert interpretation ranges after SME form finalization]	[TBD after evaluation data: complete after SME instrument finalization]

The final manuscript should use the same interpretation ranges consistently across Chapter 4, Chapter 5, and Chapter 6. No weighted mean, rank, descriptive interpretation, or acceptability claim should be added until the evaluation data and scoring procedure are complete.

4.1.4. Respondent Profile and Platform Exposure

Before presenting the outcome results, the study reports the final participant profile and the platform exposure observed during the two-week field deployment. These summaries clarify the sample used for interpretation and show whether the target platforms produced usable data.

Table 10. Participant Profile Summary

Profile Variable	Intervention Group	Control Group	Total	Remarks
Enrolled participants	[TBD after evaluation data: insert n]	[TBD after evaluation data: insert n]	[TBD after evaluation data: insert N]	[TBD after evaluation data: insert remarks]

Profile Variable	Intervention Group	Control Group	Total	Remarks
Completed Week 1	[TBD after evaluation data: insert n]	[TBD after evaluation data: insert n]	[TBD after evaluation data: insert N]	[TBD after evaluation data: insert remarks]
Completed Week 2	[TBD after evaluation data: insert n]	[TBD after evaluation data: insert n]	[TBD after evaluation data: insert N]	[TBD after evaluation data: insert remarks]
Included in primary behavioral analysis	[TBD after evaluation data: insert n]	[TBD after evaluation data: insert n]	[TBD after evaluation data: insert N]	[TBD after evaluation data: insert exclusion basis]
Included in baseline convergent association	[TBD after evaluation data: insert n]	[TBD after evaluation data: insert n]	[TBD after evaluation data: insert N]	[TBD after evaluation data: requires >= 3 reliable Week 1 sessions]

This table should be used to explain participant flow from enrollment to analysis. If the final analyzed sample is lower than the recruited sample, the interpretation should specify whether attrition, missing survey responses, device incompatibility, service interruption, or sentiment-unreliable sessions caused the reduction.

Table 11. Target Platform Use and Extraction Coverage

Platform	Participants Using Platform	Retained Sessions	Extraction Success Rate	Interpretation Notes
TikTok	[TBD after evaluation data: insert n / %]	[TBD after evaluation data: insert count]	[TBD after evaluation data: insert %]	[TBD after evaluation data: insert stability note]
Facebook Reels	[TBD after evaluation data: insert n / %]	[TBD after evaluation data: insert count]	[TBD after evaluation data: insert %]	[TBD after evaluation data: insert stability note]
Instagram Reels	[TBD after evaluation data: insert n / %]	[TBD after evaluation data: insert count]	[TBD after evaluation data: insert %]	[TBD after evaluation data: insert stability note]
Multiple-platform use	[TBD after evaluation data: insert n / %]	[TBD after evaluation data: insert count]	[TBD after evaluation data: insert %]	[TBD after evaluation data: insert cross-platform note]

The platform summary should be used to limit claims to the platforms that produced stable extraction and analyzable logs. If one target platform yields low extraction success, the final discussion should avoid generalizing equally across all target platforms.

4.1.5. Research Question 2: Week 1 to Week 2 Behavioral and Self-Report Changes

The second research question asks what short-term Week 1-to-Week 2 changes are observed in selected logged usage metrics and self-reported doomscrolling scores between the intervention group and the logging-only control group, and within the intervention group across the same period. The primary analysis uses change scores, computed as Week 2 minus Week 1, with the control group used to contextualize natural fluctuation and monitoring awareness.

Table 12. Primary Between-Group Change-Score Comparison

Outcome Variable	Analyzed n	Intervention Change Mean (SD)	Control Change Mean (SD)	Test Used	Effect Size (95% CI)	Holm- Adjusted p
Mean daily raw elapsed session duration (minutes)	[TBD after evaluation data: insert n]	[TBD after evaluation data: insert mean (SD)]	[TBD after evaluation data: insert mean (SD)]	[TBD after evaluation data: t/ Welch/ Mann-Whitney]	[TBD after evaluation data: insert d and CI]	[TBD after evaluation data: insert p]
Mean daily raw elapsed video dwell time (seconds)	[TBD after evaluation data: insert n]	[TBD after evaluation data: insert mean (SD)]	[TBD after evaluation data: insert mean (SD)]	[TBD after evaluation data: t/ Welch/ Mann-Whitney]	[TBD after evaluation data: insert d and CI]	[TBD after evaluation data: insert p]
Mean daily Negative Sentiment Density (%)	[TBD after evaluation data: insert n]	[TBD after evaluation data: insert mean (SD)]	[TBD after evaluation data: insert mean (SD)]	[TBD after evaluation data: t/ Welch/ Mann-Whitney]	[TBD after evaluation data: insert d and CI]	[TBD after evaluation data: insert p]

Outcome Variable	Analyzed n	Intervention Change Mean (SD)	Control Change Mean (SD)	Test Used	Effect Size (95% CI)	Holm-Adjusted p
Pre/Post Doomscrolling Scale score	[TBD after evaluation data: insert n]	[TBD after evaluation data: insert mean (SD)]	[TBD after evaluation data: insert mean (SD)]	[TBD after evaluation data: t/Welch/Mann-Whitney]	[TBD after evaluation data: insert d and CI]	[TBD after evaluation data: insert p]

The final interpretation should identify the direction, size, and uncertainty of each observed change. A favorable result would be discussed only if the intervention group shows a stronger desirable change than the control group, with attention to effect size and confidence interval. A weak or mixed result should be reported as evidence of limited short-term difference under the pilot conditions rather than as a failure of the broader concept.

Table 13. Secondary Within-Intervention Pre/Post Comparison

Outcome Variable	Analyzed n	Week 1 Mean (SD)	Week 2 Mean (SD)	Test Used	p-value / Effect Size
Mean daily raw elapsed session duration (minutes)	[TBD after evaluation data: insert n]	[TBD after evaluation data: insert mean (SD)]	[TBD after evaluation data: insert mean (SD)]	[TBD after evaluation data: paired t/Wilcoxon]	[TBD after evaluation data: insert p and d]
Mean daily raw elapsed video dwell time (seconds)	[TBD after evaluation data: insert n]	[TBD after evaluation data: insert mean (SD)]	[TBD after evaluation data: insert mean (SD)]	[TBD after evaluation data: paired t/Wilcoxon]	[TBD after evaluation data: insert p and d]
Mean daily Negative Sentiment Density (%)	[TBD after evaluation data: insert n]	[TBD after evaluation data: insert mean (SD)]	[TBD after evaluation data: insert mean (SD)]	[TBD after evaluation data: paired t/Wilcoxon]	[TBD after evaluation data: insert p and d]
Doomscrolling Scale score	[TBD after evaluation data: insert n]	[TBD after evaluation data: insert mean (SD)]	[TBD after evaluation data: insert mean (SD)]	[TBD after evaluation data: paired t/Wilcoxon]	[TBD after evaluation data: insert p and d]

This secondary comparison should be interpreted cautiously because it does not by itself separate the intervention effect from monitoring awareness, time variation, or other external

factors. It is retained to describe the intervention group’s observed pre/post pattern and to support comparison with the primary between-group analysis.

Table 14. Supplementary Prompt-Excluded Active-Use Metrics

Metric	Analyzed n	Raw Elapsed Result	Prompt-Excluded Result	Measurement Interpretation
Session duration	[TBD after evaluation data: insert n]	[TBD after evaluation data: insert result]	[TBD after evaluation data: insert result]	[TBD after evaluation data: explain whether prompts changed measured active-use time]
Video dwell time	[TBD after evaluation data: insert n]	[TBD after evaluation data: insert result]	[TBD after evaluation data: insert result]	[TBD after evaluation data: explain whether prompt time affected dwell estimates]

The prompt-excluded metrics should be used as a transparency check. If raw elapsed and prompt-excluded results differ, the final discussion should explain whether the difference reflects prompt interruption time rather than ordinary target-platform use.

4.1.6. Research Question 3: Baseline Convergent Association Between DSI and Self-Reported Doomscrolling

The third research question asks what baseline convergent association exists between the fixed-prior Week 1 DSI and participants’ self-reported doomscrolling scores among participants with at least three sentiment-reliable Week 1 sessions. This analysis does not validate the system as a diagnostic classifier; it checks whether the baseline composite estimator relates plausibly to self-reported doomscrolling under the pilot conditions.

Table 15. DSI Coverage and Sentiment-Reliable Session Summary

Coverage Indicator	Intervention Group	Control Group	Total	Interpretation Notes
Participants with ≥ 3 sentiment-reliable Week 1 sessions	[TBD after evaluation data: insert n]	[TBD after evaluation data: insert n]	[TBD after evaluation data: insert N_effective]	[TBD after evaluation data: insert coverage interpretation]
Week 1 sentiment-reliable sessions	[TBD after evaluation data: insert count]	[TBD after evaluation data: insert count]	[TBD after evaluation data: insert count]	[TBD after evaluation data: insert retained-count]

Coverage Indicator	Intervention Group	Control Group	Total	Interpretation Notes
				session interpretation]
Week 1 sentiment-unreliable sessions	[TBD after evaluation data: insert count]	[TBD after evaluation data: insert count]	[TBD after evaluation data: insert count]	[TBD after evaluation data: insert exclusion interpretation]
Undefined Week 1 DSI cases	[TBD after evaluation data: insert count]	[TBD after evaluation data: insert count]	[TBD after evaluation data: insert count]	[TBD after evaluation data: insert reason]

The DSI coverage table should be interpreted before the correlation result. If many participants or sessions are excluded because of sentiment-unreliable data, the final discussion should state that estimator plausibility is based on the retained analyzable subset.

Table 16. Baseline Component and Composite Correlation with Doomscrolling Scale

Baseline Variable	N_effective	Correlation Type	Coefficient	p-value	Interpretation
Mean daily session duration	[TBD after evaluation data: insert n]	[TBD after evaluation data: Pearson r / Spearman rho]	[TBD after evaluation data: insert coefficient]	[TBD after evaluation data: insert p]	[TBD after evaluation data: interpret direction and size]
Mean daily video dwell time	[TBD after evaluation data: insert n]	[TBD after evaluation data: Pearson r / Spearman rho]	[TBD after evaluation data: insert coefficient]	[TBD after evaluation data: insert p]	[TBD after evaluation data: interpret direction and size]
Mean daily NSD	[TBD after evaluation data: insert n]	[TBD after evaluation data: Pearson r / Spearman rho]	[TBD after evaluation data: insert coefficient]	[TBD after evaluation data: insert p]	[TBD after evaluation data: interpret direction and size]
Composite Week 1 DSI	[TBD after evaluation data: insert n]	[TBD after evaluation data: Pearson r / Spearman rho]	[TBD after evaluation data: insert coefficient]	[TBD after evaluation data: insert p]	[TBD after evaluation data: interpret plausibility]

The final interpretation should compare the composite DSI association against its individual components. If DSI shows a stronger and theoretically coherent association than session

duration alone, it may be discussed as preliminary estimator plausibility. If the association is weak, inconsistent, or based on low coverage, the finding should be framed as limited pilot evidence rather than as evidence of invalidity or validity.

4.1.7. ISO/IEC 25010 Software-Quality Criteria and Per-Category User Results

The fourth research question asks how users evaluate the system using ISO/IEC 25010 and TAM. The user evaluation is reported through per-category descriptive statistics, interpretation thresholds, and internal consistency estimates. The System Usability Scale is reported separately from the researcher-developed ISO/IEC 25010 items because it uses a 0 to 100 scoring procedure.

Table 17. ISO/IEC 25010 User Evaluation Summary

Quality Characteristic	Items	Mean	SD	Target	Interpretation
Functional Suitability	[TBD after evaluation data: insert item count]	[TBD after evaluation data: insert mean]	[TBD after evaluation data: insert SD]	≥ 3.50	[TBD after evaluation data: met / not met / interpret]
Performance Efficiency	[TBD after evaluation data: insert item count]	[TBD after evaluation data: insert mean]	[TBD after evaluation data: insert SD]	≥ 3.50	[TBD after evaluation data: met / not met / interpret]
Reliability	[TBD after evaluation data: insert item count]	[TBD after evaluation data: insert mean]	[TBD after evaluation data: insert SD]	≥ 3.50	[TBD after evaluation data: met / not met / interpret]

The final interpretation should identify which ISO/IEC 25010 characteristics met the study-defined favorable threshold and which did not. Results below the threshold should be discussed as areas requiring improvement, not omitted.

Table 18. System Usability Scale Result

Usability Measure	Analyzed n	Mean Score	Target	Interpretation
System Usability Scale (0-100)	[TBD after evaluation data: insert n]	[TBD after evaluation data: insert SUS mean]	≥ 70	[TBD after evaluation data: met / not met / interpret against target]

The SUS result should be interpreted using the study target of 70. If the result is below the target, the discussion should identify interface, onboarding, permission, prompt, or dashboard issues raised by the survey and open-ended feedback.

Table 19. Technology Acceptance Model User Evaluation Summary

TAM Construct	Items	Mean	SD	Target	Interpretation
Perceived Usefulness	[TBD after evaluation data: insert item count]	[TBD after evaluation data: insert mean]	[TBD after evaluation data: insert SD]	≥ 3.50	[TBD after evaluation data: met / not met / interpret]
Perceived Ease of Use	[TBD after evaluation data: insert item count]	[TBD after evaluation data: insert mean]	[TBD after evaluation data: insert SD]	≥ 3.50	[TBD after evaluation data: met / not met / interpret]

The final TAM interpretation should distinguish perceived usefulness from perceived ease of use. A high usefulness score with lower ease of use would suggest perceived value but onboarding or interface friction. A low usefulness score would require discussion of whether prompts, dashboard feedback, or estimator explanations were meaningful to participants.

4.1.8. Reliability Analysis Placeholder

Cronbach's alpha is used to estimate the internal consistency of the multi-item survey subscales. It is interpreted as a reliability check for the survey responses, not as a stand-alone validation of the system or the constructs. Because the study uses a pilot-sized sample and includes researcher-developed items, each coefficient should be interpreted cautiously and alongside item content.

Table 20. Cronbach's Alpha Results for User Survey Subscales

Subscale	Items	Cronbach's Alpha	Interpretation
Functional Suitability	[TBD after evaluation data: insert k]	[TBD after evaluation data: insert alpha]	[TBD after evaluation data: ≥ 0.70 / below threshold]
Performance Efficiency	[TBD after evaluation data: insert k]	[TBD after evaluation data: insert alpha]	[TBD after evaluation data: ≥ 0.70 / below threshold]
Reliability	[TBD after evaluation data: insert k]	[TBD after evaluation data: insert alpha]	[TBD after evaluation data: ≥ 0.70 / below threshold]
Perceived Usefulness	[TBD after evaluation data: insert k]	[TBD after evaluation data: insert alpha]	[TBD after evaluation data: ≥ 0.70 / below threshold]
Perceived Ease of Use	[TBD after evaluation data: insert k]	[TBD after evaluation data: insert alpha]	[TBD after evaluation data: ≥ 0.70 / below threshold]

The final interpretation should state which subscales reached the conventional $\alpha \geq 0.70$ benchmark. If any subscale falls below the benchmark, the discussion should note whether the issue may reflect a small pilot sample, heterogeneous item wording, or weak item coherence.

4.1.9. Algorithm Complexity Analysis Placeholder

The algorithm complexity analysis should document the final implementation path after the Android application and analysis scripts are frozen. This placeholder is retained so that computational claims are not made before the implemented routines are reviewed.

Table 21. Algorithm Complexity Analysis Placeholder

Algorithm / Procedure	Input Basis	Complexity to Report	Interpretation Notes
Accessibility event processing	[TBD after evaluation data: insert final event-processing input definition]	[TBD after evaluation data: insert time and space complexity after implementation review]	[TBD after evaluation data: insert performance implication]
Sentiment and fallback routing	[TBD after evaluation data: insert final text/no-text routing basis]	[TBD after evaluation data: insert time and space complexity after implementation review]	[TBD after evaluation data: insert device constraint implication]

Algorithm / Procedure	Input Basis	Complexity to Report	Interpretation Notes
Fuzzy risk estimation	[TBD after evaluation data: insert final variable and rule-count basis]	[TBD after evaluation data: insert time and space complexity after implementation review]	[TBD after evaluation data: insert scalability implication]
Local aggregation and export	[TBD after evaluation data: insert final stored-record and export basis]	[TBD after evaluation data: insert time and space complexity after implementation review]	[TBD after evaluation data: insert storage and privacy implication]

The final discussion should report only the complexity supported by the reviewed source code and should avoid claiming efficiency beyond the tested implementation.

4.1.10. Research Question 5: Subject Matter Expert Evaluation

The fifth research question asks how subject matter experts evaluate the system's technical design, privacy safeguards, heuristic logic, and intervention structure. The SME results are presented separately from user survey results because the SME panel evaluates technical plausibility and design coherence rather than ordinary user acceptance.

Table 22. Subject Matter Expert Evaluation Summary

Evaluation Area	Experts	Mean	SD	Target	Interpretation
Technical soundness of risk-estimation framework	[TBD after evaluation data: insert expert count]	[TBD after evaluation data: insert mean]	[TBD after evaluation data: insert SD]	≥ 4.00	[TBD after evaluation data: met / not met / interpret]
Plausibility of input ranges	[TBD after evaluation data: insert expert count]	[TBD after evaluation data: insert mean]	[TBD after evaluation data: insert SD]	≥ 4.00	[TBD after evaluation data: met / not met / interpret]
Internal coherence of fuzzy rule base	[TBD after evaluation data: insert expert count]	[TBD after evaluation data: insert mean]	[TBD after evaluation data: insert SD]	≥ 4.00	[TBD after evaluation data: met / not met / interpret]

Evaluation Area	Experts	Mean	SD	Target	Interpretation
Privacy and architecture quality	[TBD after evaluation data: insert expert count]	[TBD after evaluation data: insert mean]	[TBD after evaluation data: insert SD]	≥ 4.00	[TBD after evaluation data: met / not met / interpret]
Intervention design appropriateness	[TBD after evaluation data: insert expert count]	[TBD after evaluation data: insert mean]	[TBD after evaluation data: insert SD]	≥ 4.00	[TBD after evaluation data: met / not met / interpret]
Overall expert evaluation	[TBD after evaluation data: insert expert count]	[TBD after evaluation data: insert mean]	[TBD after evaluation data: insert SD]	≥ 4.00	[TBD after evaluation data: met / not met / interpret]

The final interpretation should report both the numerical SME ratings and the expert comments. If experts disagree, the discussion should identify whether disagreement concerns the estimator variables, fuzzy rule base, Android extraction assumptions, privacy safeguards, or intervention design.

4.1.11. Open-Ended Feedback

Open-ended responses from the post-usage survey are analyzed through light inductive coding. The purpose is to identify recurring usability concerns, perceived intervention value, privacy concerns, device or permission issues, and implementation constraints that are not fully captured by Likert-scale responses.

Table 23. Open-Ended Feedback Coding Summary

Theme / Issue Category	Frequency	Representative Concern	Design Implication
Prompt timing and interruption	[TBD after evaluation data: insert count]	[TBD after evaluation data: insert summarized feedback, not direct identifying quote]	[TBD after evaluation data: insert implication]
Permission and onboarding clarity	[TBD after evaluation data: insert count]	[TBD after evaluation data: insert summarized feedback, not direct identifying quote]	[TBD after evaluation data: insert implication]

Theme / Issue Category	Frequency	Representative Concern	Design Implication
Dashboard usefulness	[TBD after evaluation data: insert count]	[TBD after evaluation data: insert summarized feedback, not direct identifying quote]	[TBD after evaluation data: insert implication]
Privacy and trust	[TBD after evaluation data: insert count]	[TBD after evaluation data: insert summarized feedback, not direct identifying quote]	[TBD after evaluation data: insert implication]
Technical reliability	[TBD after evaluation data: insert count]	[TBD after evaluation data: insert summarized feedback, not direct identifying quote]	[TBD after evaluation data: insert implication]

The final interpretation should connect these themes to the quantitative results only after the coded feedback and survey results are available. For example, any explanation of ease-of-use, onboarding, permission, reliability, or platform-extraction concerns should be tied to verified Chapter 4 evidence.

4.1.12. Overall Evaluation Results

The overall results table consolidates the major evaluation areas without replacing the per-category tables. It should be completed only after the preceding result sections have been populated.

Table 24. Overall Results Placeholder

Evaluation Area	Evidence Source	Overall Result to Insert	Interpretation Status
Implementation and testing	Architecture matrix, black box testing, and white box testing	[TBD after evaluation data: insert overall implementation/testing result]	[TBD after evaluation data: complete after system testing]
Behavioral and self-report outcomes	Week 1 to Week 2 logs and Doomscrolling Scale scores	[TBD after evaluation data: insert overall change pattern]	[TBD after evaluation data: complete after statistical analysis]
Estimator plausibility	Week 1 DSI coverage and correlation analysis	[TBD after evaluation data: insert overall DSI plausibility result]	[TBD after evaluation data: complete after correlation analysis]
User evaluation	ISO/IEC 25010, SUS, TAM, and open-ended feedback	[TBD after evaluation data: insert overall user evaluation result]	[TBD after evaluation data:]

Evaluation Area	Evidence Source	Overall Result to Insert	Interpretation Status
			complete after survey analysis]
Expert evaluation	SME rating form and SME comments	[TBD after evaluation data: insert overall expert evaluation result]	[TBD after evaluation data: complete after SME review]

4.2. Narrative Interpretation of Results

The interpretation of results should connect the statistical and descriptive findings back to the five research questions. This section should be completed after all tables above have been populated.

The final narrative should explain the direction, magnitude, and limitations of the verified findings without introducing conclusions that are not supported by the completed tables.

[TBD after evaluation data: insert narrative interpretation after all Chapter 4 results are available]

4.3. Summary of Findings

Table 25. Summary of Findings by Research Question

RQ	Focus	Key Result to Report	Interpretation Status
1	Privacy-preserving architecture and estimation framework	[TBD after evaluation data: insert implementation and testing finding]	[TBD after evaluation data: complete after testing]
2	Week 1 to Week 2 behavioral and self-report changes	[TBD after evaluation data: insert change-score finding]	[TBD after evaluation data: complete after analysis]
3	Baseline DSI and Doomscrolling Scale association	[TBD after evaluation data: insert correlation finding]	[TBD after evaluation data: complete after analysis]
4	User evaluation through ISO/IEC 25010 and TAM	[TBD after evaluation data: insert user evaluation finding]	[TBD after evaluation data: complete after survey analysis]
5	Subject matter expert evaluation	[TBD after evaluation data: insert SME evaluation finding]	[TBD after evaluation data: complete after SME review]

For the final manuscript, this section should avoid broad claims beyond the data. If results are favorable, the interpretation should still identify the pilot scope and proxy-based nature of the estimator. If results are mixed or below threshold, the interpretation should specify which objective was unmet and what design or deployment condition may explain the result.

Chapter 5

CONCLUSION

This chapter presents the summary of findings and conclusions of the study. Because the final participant data, survey data, subject matter expert ratings, and statistical outputs have not yet been inserted in Chapter 4, the conclusion statements below are retained as completion shells. Final wording should be completed only after Chapter 4 contains verified results.

5.1. Summary of Findings

The study was guided by five research questions. The first question focused on the design and implementation of a privacy-preserving architecture and estimation framework. The second question examined short-term Week 1-to-Week 2 changes in logged usage metrics and self-reported doomscrolling scores between the intervention group and the logging-only control group, with an additional within-intervention comparison. The third question examined the baseline convergent association between the fixed-prior Week 1 DSI and participants' self-reported doomscrolling scores. The fourth question evaluated the system from the users' perspective using ISO/IEC 25010 and TAM. The fifth question obtained subject matter expert evaluation of the technical design, privacy safeguards, heuristic logic, and intervention structure.

Table 26. Summary of Findings for Completion

RQ	Research Question Focus	Finding to Insert from Chapter 4	Conclusion Direction
1	Architecture and estimation framework	[TBD after evaluation data: insert final	[TBD after evaluation data: insert whether objective was met,

RQ	Research Question Focus	Finding to Insert from Chapter 4	Conclusion Direction
		implementation/testing finding]	partially met, or not met]
2	Week 1 to Week 2 behavioral and self-report changes	[TBD after evaluation data: insert final change-score and pre/post finding]	[TBD after evaluation data: insert observed short-term difference interpretation]
3	Baseline DSI and Doomscrolling Scale association	[TBD after evaluation data: insert final correlation and coverage finding]	[TBD after evaluation data: insert estimator plausibility interpretation]
4	User ISO/IEC 25010 and TAM evaluation	[TBD after evaluation data: insert final survey and SUS finding]	[TBD after evaluation data: insert user acceptability interpretation]
5	SME evaluation	[TBD after evaluation data: insert final expert-rating and feedback finding]	[TBD after evaluation data: insert expert appraisal interpretation]

The completed summary should briefly synthesize the main findings rather than repeat every table from Chapter 4. It should identify whether the system met the software-development objective, whether short-term behavioral differences were observed, whether the DSI showed baseline plausibility, and how users and experts evaluated the system.

5.2. Conclusions Based on Research Objectives and Questions

The conclusions should be finalized only after the Chapter 4 results are complete. Each conclusion must be tied to a verified finding, and no conclusion should state that the system was effective, acceptable, reliable, or statistically supported unless the completed data support that claim.

The final conclusion for the first research question should state whether the implemented system architecture, fuzzy inference design, fallback behavior, and privacy safeguards were completed and verified. The conclusion should also specify any implementation constraints discovered during testing, such as target-app UI changes, device compatibility issues, or

Accessibility Service screenshot/API compatibility limitations. [TBD after evaluation data: insert final RQ1 conclusion]

The final conclusion for the second research question should state whether the intervention group showed meaningful Week 1-to-Week 2 changes compared with the logging-only control group. The wording should distinguish between observed pilot differences and long-term intervention effectiveness. [TBD after evaluation data: insert final RQ2 conclusion]

The final conclusion for the third research question should state whether the fixed-prior Week 1 DSI showed a theoretically coherent association with the short-form Doomscrolling Scale among participants with sufficient sentiment-reliable sessions. This conclusion should be framed as estimator plausibility only, not diagnostic validation. [TBD after evaluation data: insert final RQ3 conclusion]

The final conclusion for the fourth research question should state how users evaluated the system across ISO/IEC 25010, SUS, and TAM constructs. If any construct falls below the study-defined threshold, the conclusion should identify it clearly and connect it to improvement needs. [TBD after evaluation data: insert final RQ4 conclusion]

The final conclusion for the fifth research question should state how subject matter experts evaluated the technical design, privacy safeguards, heuristic logic, and intervention structure. If expert feedback identifies weaknesses in the rule base, variable thresholds, or intervention timing, the conclusion should acknowledge these as refinement priorities. [TBD after evaluation data: insert final RQ5 conclusion]

The overall conclusion should remain bounded by the study design. The study is a pilot-scale software evaluation, not a clinical trial, and the system estimates risk from observable proxies rather than diagnosing doomscrolling or mental-health conditions. [TBD after evaluation data: insert overall conclusion after Chapter 4 is complete]

Chapter 6

RECOMMENDATIONS

This chapter presents recommendations for system improvement, future research, and future implementation or deployment. Because the Chapter 4 evaluation results have not yet been completed, all recommendations are retained as placeholders and should be finalized only after the weakest evaluation categories, implementation constraints, user feedback, and expert feedback are known.

6.1. Recommendations for System Improvement

System-improvement recommendations should be tied directly to verified implementation results, ISO/IEC 25010 ratings, SUS results, TAM results, open-ended feedback, and SME evaluation findings.

Table 27. System Improvement Recommendations for Completion

Improvement Area	Condition to Check in Chapter 4	Recommendation to Finalize
Platform extraction robustness	[TBD after evaluation data: insert target-app extraction issues, if any]	[TBD after evaluation data: insert system-improvement recommendation]
Android compatibility and performance	[TBD after evaluation data: insert device, OS, background-process, or performance issue, if any]	[TBD after evaluation data: insert compatibility or optimization recommendation]
Sentiment and fallback handling	[TBD after evaluation data: insert NSD coverage, no-text fallback, or code-mixed sentiment limitation, if any]	[TBD after evaluation data: insert estimator refinement recommendation]
Prompt timing and burden management	[TBD after evaluation data: insert prompt intrusiveness, timing, or cooldown feedback, if any]	[TBD after evaluation data: insert adaptive-prompt improvement recommendation]
Dashboard and user-facing transparency	[TBD after evaluation data: insert usability, usefulness, or explanation weakness, if any]	[TBD after evaluation data: insert dashboard or explanation improvement recommendation]

The final recommendations in this section should prioritize the lowest-rated or most constrained areas reported in Chapter 4. [TBD after evaluation data: insert prioritized system-improvement recommendations after analysis of weakest evaluation categories]

6.2. Recommendations for Future Development

Future-development recommendations should identify feature, algorithm, interface, or architecture improvements that are justified by the completed Chapter 4 results.

Table 28. Future Development Recommendations for Completion

Development Area	Condition to Check in Chapter 4	Recommendation to Finalize
Risk-estimation model refinement	[TBD after evaluation data: insert DSI, NSD, fuzzy-rule, or fallback weakness, if any]	[TBD after evaluation data: insert future model-development recommendation]
Platform support expansion	[TBD after evaluation data: insert target-platform coverage and extraction-stability limitations]	[TBD after evaluation data: insert future platform-support recommendation]
User interface and dashboard iteration	[TBD after evaluation data: insert SUS, TAM, and open-ended usability findings]	[TBD after evaluation data: insert future UI development recommendation]
Data export and reporting workflow	[TBD after evaluation data: insert researcher, participant, or SME feedback on reporting needs]	[TBD after evaluation data: insert future reporting-feature recommendation]

The final recommendations in this section should remain tied to verified system-development needs and should not assume evaluation outcomes that are not yet available. [TBD after evaluation data: insert future-development recommendations after final Chapter 4 findings]

6.3. Recommendations for Future Researchers

Future-research recommendations should be grounded in the final study limitations, sample coverage, measurement reliability, and statistical results.

Table 29. Future Research Recommendations for Completion

Research Area	Condition to Check in Chapter 4	Recommendation to Finalize
Sample size and group balance	[TBD after evaluation data: insert analyzed sample, attrition, and group-balance limitations]	[TBD after evaluation data: insert future sample recommendation]
Deployment duration	[TBD after evaluation data: insert Week 1 to Week 2 result pattern and pilot-window limitation]	[TBD after evaluation data: insert future deployment-duration recommendation]
Validation anchors	[TBD after evaluation data: insert DSI correlation, coverage, and self-report limitation]	[TBD after evaluation data: insert future validation recommendation]
Survey reliability	[TBD after evaluation data: insert Cronbach's Alpha and item-coherence results]	[TBD after evaluation data: insert instrument refinement recommendation]
Qualitative feedback	[TBD after evaluation data: insert coded user and SME feedback themes]	[TBD after evaluation data: insert future mixed-methods recommendation]

The final recommendations in this section should avoid overstating the study's evidence and should identify what future researchers need to verify next. [TBD after evaluation data: insert future-research recommendations after final Chapter 4 findings]

6.4. Recommendations for Future Implementation and Deployment

Future implementation and deployment recommendations should be based on verified deployment constraints, privacy considerations, user feedback, and expert review.

Table 30. Future Implementation and Deployment Recommendations for Completion

Deployment Area	Condition to Check in Chapter 4	Recommendation to Finalize
Privacy and data governance	[TBD after evaluation data: insert privacy-safeguard verification and SME privacy feedback]	[TBD after evaluation data: insert deployment privacy recommendation]
Participant onboarding	[TBD after evaluation data: insert onboarding, permission, or setup feedback]	[TBD after evaluation data: insert onboarding recommendation]
Monitoring and support	[TBD after evaluation data: insert service reliability, support, or troubleshooting issues]	[TBD after evaluation data: insert monitoring/support recommendation]
Institutional or classroom deployment	[TBD after evaluation data: insert acceptability,	[TBD after evaluation data: insert implementation-context recommendation]

Deployment Area	Condition to Check in Chapter 4	Recommendation to Finalize
	usefulness, and ethical constraints]	
Maintenance and platform updates	[TBD after evaluation data: insert target-platform change or compatibility issue]	[TBD after evaluation data: insert maintenance recommendation]

The final recommendations in this section should describe how the system may be responsibly improved or deployed after the pilot findings are known. [TBD after evaluation data: insert future implementation/deployment recommendations after final Chapter 4 findings]

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APPENDICES

APPENDIX A

Complete 27-Rule Fuzzy Inference Base

The following table presents the complete 27-rule inference base used by the three-input fuzzy engine (Video Dwell Time, Negative Sentiment Density, Session Duration). The 27 rows arise from exhaustive coverage of the $3 \times 3 \times 3$ input-state combinations. Their Safe/Warning/Critical outputs are fixed study priors intended for transparent heuristic inference rather than empirically calibrated probabilities or clinical thresholds. The assignment logic is conservative: Safe is reserved for low-baseline or isolated single-signal states, Warning is the default for mixed or moderate states, and Critical is reserved for combinations where the main risk axes converge under the fixed study priors.

Table 31. Full 27-Rule Fuzzy Inference Base

No.	Video Dwell Time	Negative Sentiment Density	Session Duration	Risk Level
1	Low	Low	Low	Safe
2	Low	Low	Medium	Safe
3	Low	Low	High	Warning
4	Low	Medium	Low	Safe
5	Low	Medium	Medium	Warning
6	Low	Medium	High	Warning
7	Low	High	Low	Warning
8	Low	High	Medium	Warning
9	Low	High	High	Critical
10	Medium	Low	Low	Safe
11	Medium	Low	Medium	Warning
12	Medium	Low	High	Warning
13	Medium	Medium	Low	Warning
14	Medium	Medium	Medium	Warning
15	Medium	Medium	High	Critical
16	Medium	High	Low	Warning

Table 6. Full 27-Rule Fuzzy Inference Base (continued)

No.	Video Dwell Time	Negative Sentiment Density	Session Duration	Risk Level
17	Medium	High	Medium	Critical
18	Medium	High	High	Critical
19	High	Low	Low	Safe
20	High	Low	Medium	Warning
21	High	Low	High	Warning
22	High	Medium	Low	Warning
23	High	Medium	Medium	Critical
24	High	Medium	High	Critical
25	High	High	Low	Critical
26	High	High	Medium	Critical
27	High	High	High	Critical

Interpretive note: the table follows a mostly non-decreasing assignment pattern in Session Duration and Negative Sentiment Density, while using Video Dwell Time as a contextual modifier rather than a standalone high-risk trigger. One deliberate exception is retained as a precautionary prototype rule for rapid negative-content chaining. Low dwell combined with high Negative Sentiment Density and high session duration is classified as **Critical** because rapid movement does not reduce concern when the session is both prolonged and clearly negative-dominant overall. By contrast, Rule 19 (High Dwell, Low NSD, Low Session Duration) stays **Safe** because sustained attention to one item, without prolonged use or negative-dominant exposure, is not enough by itself to imply doomscrolling. This keeps dwell time as a modifier, not the main driver.

Operationally, the 27 consequents follow a compact assignment grammar rather than 27 unrelated judgments: no rule is **Critical** when both Session Duration and Negative Sentiment Density are Low; duration alone does not produce **Critical** when NSD remains Low; High NSD reaches **Critical** only when reinforced by another non-low axis; High Dwell Time alone is insufficient for **Critical**; and **Warning** remains the default mixed class when the signals are concerning but not yet strongly convergent.

Read in groups, the table has three layers. Rules 1, 2, 4, 10, and 19 remain **Safe** because at least two axes stay low. Rules 3, 5-8, 11-14, 16, and 20-22 remain **Warning** because they show mixed evidence without strong convergence between duration and negativity. Rules 9, 15, 17-18, and 23-27 become **Critical** because prolonged use and negative exposure rise together, or because High NSD is reinforced by High Dwell Time even before the session becomes long.

Within that Critical layer, the strongest challenge cases are still explainable. Rule 15 becomes **Critical** because a High-duration session with both other axes already at Medium no longer reflects time alone. Rule 23 becomes **Critical** because High Dwell Time, Medium NSD, and Medium Session Duration already show three non-low signals in the same direction. Rule 25 also remains **Critical** because a short session can still contain concentrated negative immersion when the viewed items are mostly negative and each item holds attention for longer.

APPENDIX B

Fallback Rules and Trigger Thresholds

Unless otherwise noted, the following thresholds are fixed study priors selected for reproducibility and conservative intervention behavior. They are design parameters, not empirically validated clinical or population cutoffs. Because the fallback engine retains two inputs with three linguistic levels each, exhaustive coverage requires $3^2 = 9$ rules.

In the current field-tested artifact, the two-input fallback rule base below is reserved for **Sentiment-Unreliable** sessions, specifically sessions where negativity cannot be resolved reliably after the available text or no-text path is attempted. This includes sessions whose extracted text is at least half unresolved vocabulary (study-defined unreliability trigger: session OOV ratio $\geq 50\%$) and sessions where the no-text VLM path cannot produce a stable item label because `AccessibilityService.takeScreenshot` is unavailable, screenshot capture fails, or VLM inference fails. The $\geq 50\%$ value is not treated as a validated linguistic cutoff. It is a mathematically derived majority-representativeness screen: once unresolved tokens reach half of the valid tokens in a session, the recognized tokens no longer form the majority of the lexical content available to the text path. This is therefore used as a practical safeguard for noisy code-mixed text with spelling variation, lexical borrowing, and other textual variation rather than as a claimed language norm (Mohammed & Prasad, 2023; Perera & Caldera, 2024; Nazir et al., 2026; Khan et al., 2025). In those cases, the system degrades to a two-input fallback rule base (Video Dwell Time + Session Duration):

Table 32. Fallback 9-Rule Base (Sentiment-Unreliable Sessions)

No.	Video Dwell Time	Session Duration	Risk Level
1	Low	Low	Safe
2	Low	Medium	Safe
3	Low	High	Warning
4	Medium	Low	Safe
5	Medium	Medium	Warning

No.	Video Dwell Time	Session Duration	Risk Level
6	Medium	High	Warning
7	High	Low	Warning

Table 7. Fallback 9-Rule Base (Sentiment-Unreliable Sessions) (continued)

No.	Video Dwell Time	Session Duration	Risk Level
8	High	Medium	Critical
9	High	High	Critical

In Week 1 and in any participant without locked personalization, both fallback inputs use the global memberships reported in Appendix C. Once Week 2 personalization is locked, the fallback engine retains the same 9-rule consequents and RiskScore cutoffs but replaces only the Session Duration memberships with participant-specific Week 1 duration quantiles; Video Dwell Time remains fixed. This is because the fallback mode reuses the same live prompt engine’s duration memberships while dropping only the sentiment input.

RiskScore cutoffs used across both full and fallback modes (equal-width reporting categories on the normalized 0-100 scale):

- Safe: 0.00 to 33.32
- Warning: 33.33 to 66.66
- Critical: 66.67 to 100

Prompt mapping (study-specified defaults derived from the reporting bands):

- Level 1: RiskScore 33.33 to 49.99
- Level 2: RiskScore 50.00 to 66.66
- Level 3: RiskScore 66.67 to 100 (disabled in fallback mode)
- The Warning band is split at its midpoint 50.00 so lower- and upper-Warning states receive different prompt intensities without introducing extra off-band cutoffs; Level 3 is reserved for the Critical band.
- Cooldown: fixed 15-minute prototype cooldown between prompts, intentionally reusing the same literature-bounded 15-minute live gate as the minimum re-

eligibility interval so the system does not introduce a second unsupported minute constant; a repeat prompt therefore requires another prolonged interval of continued use. Recent JITAI work supports managing timing, receptivity, and dose, but not this exact interval (Teepe et al., 2021; Wang et al., 2023; Fiedler et al., 2024; Hsu et al., 2025; van Genugten et al., 2025)

APPENDIX C

Core Algorithmic Logic and Reproducibility Specifications

The field-tested artifact separates fixed analytic priors from Week 2 live-prompt personalization. Fixed global memberships are retained for week-level DSI computation across both study weeks so baseline and intervention summaries remain comparable. A separate live prompt engine is then allowed to replace only the Session Duration and NSD memberships after Week 1; Video Dwell Time remains fixed.

Fixed Analytic Priors (used for week-level DSI across both study weeks and as the Week 1 live default): Where

$$\text{tri}(a, b, c)$$

is the standard triangular membership function, the study uses the following fixed memberships:

$$\text{Low}_{\text{Dwell}} = \text{tri}(0, 0, 5)$$

$$\text{Medium}_{\text{Dwell}} = \text{tri}(4, 12, 20)$$

$$\text{High}_{\text{Dwell}} = \text{tri}(15, 30, 30)$$

$$\text{Low}_{\text{NSD}} = \text{tri}(0, 0, 33)$$

$$\text{Medium}_{\text{NSD}} = \text{tri}(17, 50, 83)$$

$$\text{High}_{\text{NSD}} = \text{tri}(67, 100, 100)$$

$$\text{Low}_{\text{Duration}} = \text{tri}(0, 0, 10)$$

$$\text{Medium}_{\text{Duration}} = \text{tri}(8, 15, 20)$$

$$\text{High}_{\text{Duration}} = \text{tri}(15, 40, 40)$$

NSD values are clipped to $[0, 100]$. These fixed memberships govern week-level analytic DSI computation across both study weeks. In Week 1 they also act as the live default priors before any participant-specific personalization is available.

Where recent studies provide direct numeric anchors, only those anchors are treated as literature-bounded values: 5 s and 30 s for dwell time, 10 min, 15 min, and 40 min for session duration, and the normalized NSD/output partitions derived mathematically from the 0-100 scales. The remaining interior overlap points in the fixed priors are analytic

smoothing values used to preserve overlap between adjacent triangular sets and prevent the engine from collapsing into hard step thresholds. They are therefore treated as transition parameters and included in the sensitivity analysis rather than defended as standalone validated behavioral cutoffs.

Week 2 Prompt Personalization (implemented in the live prompt engine only): For participant u , let the Week 1 sentiment-reliable-session values for personalized variable X (Session Duration or NSD) be $x_{u,1}, x_{u,2}, \dots, x_{u,n_u}$. The prompt engine computes ordered quantiles:

$$Q_{25,u,X}, Q_{50,u,X}, Q_{75,u,X}, Q_{95,u,X}$$

The personalized Week 2 memberships are then:

$$\text{Low}_{X,\text{prompt},u} = \text{tri}(0, 0, Q_{50,u,X})$$

$$\text{Medium}_{X,\text{prompt},u} = \text{tri}(Q_{25,u,X}, Q_{50,u,X}, Q_{75,u,X})$$

$$\text{High}_{X,\text{prompt},u} = \text{tri}(Q_{50,u,X}, Q_{75,u,X}, Q_{95,u,X})$$

These quantiles are not treated as validated behavioral cutoffs. They are nonparametric order-statistic anchors chosen to map a three-term Low/Medium/High partition onto each participant's Week 1 distribution with the smallest transparent summary set: $\frac{Q_{25}}{Q_{75}}$ summarize the interquartile core, while Q_{95} caps the High set without letting one raw maximum set the upper endpoint. Video Dwell Time remains fixed to the global priors above. If $n_u < 10$ sentiment-reliable Week 1 sessions, the default priors are retained for that participant rather than locking personalized bounds. This 10-session minimum remains a numerical sufficiency safeguard rather than a validated population cutoff: below $n_u = 10$, the empirical Q_{95} is governed too heavily by the single largest observation or a nearly maximal order statistic, so the personalized High endpoint becomes too sensitive to one extreme session. Recent personalized intervention-criteria work supports deriving live criteria from a first-week baseline, but recent JITAI reviews also note that empirical decision rules and points remain underdeveloped, which is why the fielded artifact prefers participant-specific Week 2 criteria only when enough baseline data exist (Ikegaya et al., 2025; Hsu et al., 2025; van

Genugten et al., 2025; Elmer et al., 2025). Because fallback prompting uses the same live prompt engine, personalized Session Duration bounds also carry into fallback mode when personalization is available; only the rule base and prompt cap differ.

Risk Score Category Cutoffs (0-100 scale):

- Safe: $0 \leq \text{RiskScore} < 33.33$
- Warning: $33.33 \leq \text{RiskScore} < 66.67$
- Critical: $66.67 \leq \text{RiskScore} \leq 100$

Output Centers Used in CoG Aggregation:

- Safe center: 16.67
- Warning center: 50
- Critical center: 83.33

These fixed centers are the band midpoints of the equal-width thirds above, so they are mathematically derived ordinal anchors rather than empirical risk probabilities. This removes arbitrary-looking output constants while preserving smooth interpolation across mixed-rule activations before the final category label is assigned.

Intervention Mapping and Cooldown:

- No intervention when $\text{RiskScore} < 33.33$
- Level 1 (Awareness Notification): $33.33 \leq \text{RiskScore} < 50$
- Level 2 (Pause Prompt): $50 \leq \text{RiskScore} < 66.67$
- Level 3 (Breathing Exercise): $66.67 \leq \text{RiskScore} \leq 100$
- The Warning band is split at its midpoint 50.00 so lower- and upper-Warning states receive different prompt intensities without introducing extra off-band cutoffs; Level 3 is reserved for the Critical band.
- Fixed cooldown: minimum 15-minute prototype interval between any two prompts, intentionally reusing the same literature-bounded 15-minute live gate as

the minimum re-eligibility interval so the system does not introduce a second unsupported minute constant; a repeat prompt therefore requires another prolonged interval of continued use. Recent JITAI work supports managing timing, receptivity, and dose, but not this exact interval (Teepe et al., 2021; Wang et al., 2023; Fiedler et al., 2024; Hsu et al., 2025; van Genugten et al., 2025)

When the 2-input behavioral fallback mode is active (Sentiment-Unreliable), Level 3 is disabled by safety lock and the maximum permitted prompt is Level 2.

APPENDIX D

Short-Form Doomscrolling Scale Administration Sheet

The study uses the short-form 4-item **Doomscrolling Scale**, corresponding to Items 1, 2, 10, and 12 of the original 15-item instrument developed by Sharma et al. (2022) and later confirmed in short-form use by Satici et al. (2023). To align self-report with each study week, the instrument is administered with a 7-day recall instruction stem. This modification narrows the recall window only; it does not create a separately validated weekly state instrument.

Participant Instructions: “In the past week, please indicate how strongly you agree with each statement.”

Response Format:

- 1 = Strongly disagree
- 2 = Disagree
- 3 = Somewhat disagree
- 4 = Neither agree nor disagree
- 5 = Somewhat agree
- 6 = Agree
- 7 = Strongly agree

Items:

1. I feel an urge to seek bad news on social media, more and more often.
2. I lose track of time when I read bad news on social media.
3. I find myself continuously browsing negative news.
4. I feel like I am addicted to negative news.

Scoring Note: Item responses are summed or averaged to produce a single doomscrolling score, with higher values indicating greater self-reported doomscrolling during the recalled 7-day period. The 4-item short form contains no reverse-coded items.